

Chapter 7 Wall Covering

Section R701 General

R701.1 Application

The provisions of this chapter shall control the design and construction of the interior and exterior wall covering for buildings.

Exception: Buildings and structures located within the High-Velocity Hurricane Zone shall comply with the provisions of Chapter 44.

R701.2 Installation

Products sensitive to adverse weather shall not be installed until adequate weather protection for the installation is provided. Exterior sheathing shall be dry before applying exterior cover.

Section R702 Interior Covering

R702.1 General

Interior coverings or wall finishes shall be installed in accordance with this chapter and Table R702.1(1), Table R702.1(2), Table R702.1(3) and Table R702.3.5. Interior masonry veneer shall comply with the requirements of Section R703.7.1 for support and Section R703.7.4 for anchorage, except an airspace is not required. Interior finishes and materials shall conform to the flame spread and smoke-development requirements of Section R302.9.

TABLE R702.1(1)
 THICKNESS OF PLASTER

PLASTER BASE	FINISHED THICKNESS OF PLASTER FROM FACE OF LATH, MASONRY, CONCRETE (inches)	
	Gypsum Plaster	Cement Plaster
Expanded metal lath	5⁄8, minimum ^a	5⁄8, minimum ^a
Wire lath	5⁄8, minimum ^a	3⁄4, minimum (interior) ^b
		7⁄8, minimum (exterior) ^b
Gypsum lath ^g	1⁄2, minimum	3⁄4, minimum (interior) ^b
Masonry walls ^c	1⁄2, minimum	1⁄2, minimum
Monolithic concrete walls ^{c, d}	5⁄8, maximum	7⁄8, maximum
Monolithic concrete ceilings ^{c, d}	3⁄8, maximum ^e	1⁄2, maximum
Gypsum veneer base ^{f, g}	1⁄16, minimum	3⁄4, minimum (interior) ^b
Gypsum sheathing ^g	—	3⁄4, minimum (interior) ^b
		7⁄8, minimum (exterior) ^b

For SI: 1 inch = 25.4 mm.

- a. Where measured from back plane of expanded metal lath, exclusive of ribs, or self-furring lath, plaster thickness shall be 3⁄4 inch minimum.
- b. Where measured from face of support or backing.
- c. Because masonry and concrete surfaces vary in plane, thickness of plaster need not be uniform.
- d. Where applied over a liquid bonding agent, finish coat shall be permitted to be applied directly to concrete surface.
- e. Approved acoustical plaster shall be permitted to be applied directly to concrete or over base coat plaster, beyond the maximum plaster thickness shown.
- f. Attachment shall be in accordance with Table R702.3.5.
- g. Where gypsum board is used as a base for cement plaster, a water-resistive barrier complying with Section R703.2 shall be provided.

TABLE R702.1(2)
 GYPSUM PLASTER PROPORTIONS^a

NUMBER	COAT	PLASTER BASE OR LATH	MAXIMUM VOLUME AGGREGATE PER 100 POUNDS NEAT PLASTER ^b (cubic feet)	
			Damp Loose Sand ^a	Perlite or Vermiculite ^c
Two-coat work	Base coat	Gypsum lath	2.5	2
	Base coat	Masonry	3	3
Three-coat work	First coat	Lath	2 ^d	2

	Second coat	Lath	3 ^d	2 ^e
	First and second coats	Masonry	3	3

For SI: 1 inch = 25.4 mm, 1 cubic foot = 0.0283 m³, 1 pound = 0.454 kg.

- Wood-fibered gypsum plaster shall be mixed in the proportions of 100 pounds of gypsum to not more than 1 cubic foot of sand where applied on masonry or concrete.
- Where determining the amount of aggregate in set plaster, a tolerance of 10 percent shall be allowed.
- Combinations of sand and lightweight aggregate shall be permitted to be used, provided the volume and weight relationship of the combined aggregate to gypsum plaster is maintained.
- If used for both first and second coats, the volume of aggregate shall be permitted to be 2.5 cubic feet.
- Where plaster is 1 inch or more in total thickness, the proportions for the second coat may be increased to 3 cubic feet.

TABLE R702.1(3)

CEMENT PLASTER PROPORTIONS, PARTS BY VOLUME

COAT	CEMENT PLASTER TYPE	CEMENTITIOUS MATERIALS				VOLUME OF AGGREGATE PER SUM OF SEPARATE VOLUMES OF CEMENTITIOUS MATERIALS ^b
		Portland Cement Type I, II or III or Blended Hydraulic Cement Type IP, IS(S<70), IL or IT(S<70); or Hydraulic Cement Type GU, HE, MS, HS or MH	Plastic Cement	Masonry Cement Type M, S or N	Lime	
First	Portland or blended	1			3/4 - 1 1/2 ^a	2 1/2 - 4
	Masonry				1	2 1/2 - 4
	Plastic		1			2 1/2 - 4
Second	Portland or blended	1			3/4 - 1 1/2	3 - 5
	Masonry			1		3 - 5
	Plastic		1			3 - 5
Finish	Portland or blended	1			3/4 - 2	1 1/2 - 3
	Masonry			1		1 1/2 - 3
	Plastic		1			1 1/2 - 3

For SI: 1 inch = 25.4 mm, 1 pound = 0.454 kg.

- Lime by volume of 0 to 3/4 shall be used where the plaster will be placed over low-absorption surfaces such as dense clay tile or brick.
- The same or greater sand proportion shall be used in the second coat than used in the first coat.

R702.2 Interior Plaster

R702.2.1 Gypsum Plaster

Gypsum plaster materials shall conform to ASTM C5, C22, C28, C35, C59, C61, C587, C631, C847, C 933, C1032 and C1047, and shall be installed or applied in compliance with ASTM C843 and C844. Gypsum lath or gypsum base for veneer plaster shall conform to ASTM C1396. Plaster shall be not less than three coats where applied over metal lath and not less than two coats where applied over other bases permitted by this section, except that veneer plaster shall be applied in one coat not to exceed 3/16 inch (4.76 mm) thickness, provided the total thickness is in accordance with Table R702.1(1).

R702.2.2 Cement Plaster

Cement plaster materials shall conform to ASTM C91 (Type M, S or N), C150 (Type I, II and III), C595 [Type IP, I (PM), IS and I (SM)], C847, C897, C926, C933, C1032, C1047 and C 1328, and shall be installed or applied in compliance with ASTM C1063. Gypsum lath shall conform to ASTM C1396. Plaster shall be not less than three coats where applied over metal lath and not less than two coats where applied over other bases permitted by this section, except that veneer plaster shall be applied in one coat not to exceed 3/16 inch (4.76 mm) thickness, provided the total thickness is in accordance with Table R702.1(1).

R702.2.2.1 Application

Each coat shall be kept in a moist condition for not less than 24 hours prior to application of the next coat.

Exception: Applications installed in accordance with ASTM C926.

R702.2.2.2 Curing

[illegible]

$\frac{3}{8}$	Ceiling ^d	Perpendicular	16	16	16	Same as above for $\frac{3}{8}$ " gypsum board and gypsum panel products.
	Wall	Either direction	16	16	24	
$\frac{1}{2}$ or $\frac{5}{8}$	Ceiling	Either direction	16	16	16	Same as above for $\frac{1}{2}$ " and $\frac{5}{8}$ " gypsum board and gypsum panel products, respectively.
	Ceiling ^d	Perpendicular	24	12	16	
	Wall	Either direction	24	16	24	
Two $\frac{3}{8}$ layers	Ceiling	Perpendicular	16	16	16	Base ply nailed as above for $\frac{1}{2}$ " gypsum board and gypsum panel products; face ply installed with adhesive.
	Wall	Either direction	24	24	24	

For SI: 1 inch = 25.4 mm.

- For application without adhesive, a pair of nails spaced not less than 2 inches apart or more than $2\frac{1}{2}$ inches apart shall be permitted to be used with the pair of nails spaced 12 inches on center.
- Screws shall be in accordance with Section R702.3.5.1. Screws for attaching gypsum board or gypsum panel products to structural insulated panels shall penetrate the wood structural panel facing not less than $\frac{7}{16}$ inch.
- Where cold-formed steel framing is used with a clinching design to receive nails by two edges of metal, the nails shall be not less than $\frac{5}{8}$ inch longer than the gypsum board or gypsum panel product thickness and shall have ringed shanks. Where the cold-formed steel framing has a nailing groove formed to receive the nails, the nails shall have barbed shanks or be 0.086 inch diameter, $1\frac{5}{8}$ inches long, $\frac{15}{64}$ -inch head for $\frac{1}{2}$ -inch gypsum board or gypsum panel product; and 0.099 inch diameter, $1\frac{7}{8}$ inches long, $\frac{15}{64}$ -inch head for $\frac{5}{8}$ -inch gypsum board or gypsum panel product.
- Three-eighths-inch-thick single-ply gypsum board or gypsum panel product shall not be used on a ceiling where a water-based textured finish is to be applied, or where it will be required to support insulation above a ceiling. On ceiling applications to receive a water-based texture material, either hand or spray applied, the gypsum board or gypsum panel product shall be applied perpendicular to framing. Where applying a water-based texture material, the minimum gypsum board thickness shall be increased from $\frac{3}{8}$ inch to $\frac{1}{2}$ inch for 16-inch on center framing, and from $\frac{1}{2}$ inch to $\frac{5}{8}$ inch for 24-inch on center framing or $\frac{1}{2}$ -inch sag-resistant gypsum ceiling board shall be used.

R702.3.5.1 Screw Fastening

Screws for attaching gypsum board and gypsum panel products to wood framing shall be Type W or Type S in accordance with ASTM C1002 and shall penetrate the wood not less than $\frac{5}{8}$ inch (15.9 mm). Gypsum board and gypsum panel products shall be attached to cold-formed steel framing with minimum No. 6 screws. Screws for attaching gypsum board and gypsum panel products to cold-formed steel framing less than 0.033 inch (1 mm) thick shall be Type S in accordance with ASTM C1002 or bugle head style in accordance with ASTM C1513 and shall penetrate the steel not less than $\frac{3}{8}$ inch (9.5 mm). Screws for attaching gypsum board and gypsum panel products to cold-formed steel framing 0.033 inch to 0.112 inch (1 mm to 3 mm) thick shall be in accordance with ASTM C954 or bugle head style in accordance with ASTM C1513. Screws for attaching gypsum board and gypsum panel products to structural insulated panels shall penetrate the wood structural panel facing not less than $\frac{7}{16}$ inch (11.1 mm).

R702.3.6 Horizontal Gypsum Board Diaphragm Ceilings

Gypsum board and gypsum panel products shall be permitted on wood joists to create a horizontal *diaphragm* in accordance with Table R702.3.6. Gypsum board and gypsum panel products shall be installed perpendicular to ceiling framing members. End joints of adjacent courses of board and panels shall not occur on the same joist. The maximum allowable *diaphragm* proportions shall be $1\frac{1}{2}$:1 between shear resisting elements. Rotation or cantilever conditions shall not be permitted. Gypsum board or gypsum panel products shall not be used in *diaphragm* ceilings to resist lateral forces imposed by masonry or concrete construction. Perimeter edges shall be blocked using wood members not less than 2-inch by 6-inch (51 mm by 152 mm) nominal dimension. Blocking material shall be installed flat over the top plate of the wall to provide a nailing surface not less than 2 inches (51 mm) in width for the attachment of the gypsum board or gypsum panel product.

TABLE R702.3.6

ALLOWABLE (ASD) SHEAR CAPACITY FOR HORIZONTAL WOOD-FRAMED GYPSUM BOARD DIAPHRAGM CEILING ASSEMBLIES

MATERIAL	THICKNESS OF MATERIAL (min.) (inch)	SPACING OF FRAMING MEMBERS (max.) (inch)	SHEAR VALUE ^{a, b} (plf of ceiling)	MINIMUM FASTENER SIZE ^{c, d}
Gypsum board or gypsum panel product	$\frac{1}{2}$	16 o.c.	90	5d cooler or wallboard nail; $1\frac{5}{8}$ -inch long; 0.086-inch shank; $\frac{15}{64}$ -inch head
Gypsum board or gypsum panel product	$\frac{1}{2}$	24 o.c.	70	5d cooler or wallboard nail; $1\frac{5}{8}$ -inch long; 0.086-inch shank; $\frac{15}{64}$ -inch head

For SI: 1 inch = 25.4 mm, 1 pound per linear foot = 1.488 kg/m.

- Values are not cumulative with other horizontal diaphragm values and are for short-term loading caused by wind loading. Values shall be reduced 25 percent for normal loading.
- Reserved.
- $1\frac{1}{4}$ -inch, No. 6 Type S or W screws shall be permitted to be substituted for the listed nails.

d. Fasteners shall be spaced not more than 7 inches on center at all supports, including perimeter blocking, and not less than 3/8 inch from the edges and ends of the gypsum board.

R702.3.7 Water-Resistant Gypsum Backing Board

Gypsum board used as the base or backer for adhesive application of ceramic tile or other required nonabsorbent finish material shall conform to ASTM C1396, C1178 or C1278. Use of water-resistant gypsum backing board shall be permitted on ceilings. Water-resistant gypsum board shall not be installed over a Class I or II vapor retarder in a shower or tub compartment. Cut or exposed edges, including those at wall intersections, shall be sealed as recommended by the manufacturer.

R702.3.7.1 Limitations

Water-resistant gypsum backing board shall not be used where there will be direct exposure to water, or in areas subject to continuous high humidity.

R702.4 Ceramic Tile

R702.4.1 General

Ceramic tile surfaces shall be installed in accordance with ANSI A108.1, A108.4, A108.5, A108.6, A108.11, A118.1, A118.3, A136.1 and A137.1.

R702.4.2 Backer Boards

Materials used as backers for wall tile in tub and shower areas and wall panels in shower areas shall be of materials listed in Table R702.4.2, and installed in accordance with the manufacturer's recommendations.

TABLE R702.4.2
BACKER BOARD MATERIALS

MATERIAL	STANDARD
Glass mat gypsum backing panel	ASTM C1178
Fiber-reinforced gypsum panels	ASTM C1278
Nonasbestos fiber-cement backer board	ASTM C1288 or ISO 8336, Category C
Nonasbestos fiber mat reinforced cementitious backer units	ASTM C1325

R702.5 Other Finishes

Wood veneer paneling and hardboard paneling shall be placed on wood or cold-formed steel framing spaced not more than 16 inches (406 mm) on center. Wood veneer and hard board paneling less than 1/4-inch (6 mm) nominal thickness shall not have less than a 3/8-inch (10 mm) gypsum board or gypsum panel product backer. Wood veneer paneling not less than 1/4-inch (6 mm) nominal thickness shall conform to ANSI/HPVA HP-1. Hardboard paneling shall conform to CPA/ANSI A135.5.

R702.6 Wood Shakes and Shingles

Wood shakes and shingles shall conform to CSSB *Grading Rules for Wood Shakes and Shingles* and shall be permitted to be installed directly to the studs with maximum 24 inches (610 mm) on-center spacing.

R702.6.1 Attachment

Nails, staples or glue are permitted for attaching shakes or shingles to the wall, and attachment of the shakes or shingles directly to the surface shall be permitted provided the fasteners are appropriate for the type of wall surface material. Where nails or staples are used, two fasteners shall be provided and shall be placed so that they are covered by the course above.

R702.6.2 Furring Strips

Where furring strips are used, they shall be 1 inch by 2 inches or 1 inch by 3 inches (25 mm by 51 mm or 25 mm by 76 mm), spaced a distance on center equal to the desired exposure, and shall be attached to the wall by nailing through other wall material into the studs.

R702.7 Vapor Retarders

Vapor retarder materials shall be classified in accordance with Table R702.7(1). A vapor retarder shall be provided on the interior side of frame walls of the class indicated in Table R702.7(2), including compliance with Table R702.7(3) or Table R702.7(4) where applicable. An approved design using accepted engineering practice for hygrothermal analysis shall be permitted as an alternative. The climate zone shall be determined in accordance with Section R301.1 of the *Florida Building Code, Energy Conservation*.

Exceptions:

1. Basement walls.
2. Below-grade portion of any wall.
3. Construction where moisture accumulation, condensation, or freezing of moisture will not damage the materials.
4. A vapor retarder shall not be required in Climate Zones 1, 2 and 3.

R702.7.1 Spray Foam Plastic Insulation for Moisture Control With Class II and III Vapor Retarders

For purposes of compliance with Tables R702.7(3) and R702.7(4), spray foam with a maximum permeance of 1.5 perms at the installed thickness applied to the interior side of wood structural panels, fiberboard, insulating sheathing or gypsum shall be deemed to meet the continuous insulation moisture control requirement in accordance with one of the following conditions:

1. The spray foam R-value is equal to or greater than the specified continuous insulation R-value.

2. The combined *R*-value of the spray foam and continuous insulation is equal to or greater than the specified continuous insulation *R*-value.

TABLE R702.7(1)

VAPOR RETARDER MATERIALS AND CLASSES

CLASS	ACCEPTABLE MATERIALS
I	Sheet polyethylene, nonperforated aluminum foil, or other approved materials with a perm rating of less than or equal to 0.1.
II	Kraft-faced fiberglass batts, vapor retarder paint, or other approved materials applied in accordance with the manufacturer's installation instructions for a perm rating greater than 0.1 and less than or equal to 1.0.
III	Latex paint, enamel paint or other approved materials applied in accordance with the manufacturer's installation instructions for a perm rating of greater than 1.0 and less than or equal to 10.0.

TABLE R702.7(2)

VAPOR RETARDER OPTIONS

CLIMATE ZONE	VAPOR RETARDER CLASS		
	CLASS I ^a	CLASS II ^a	CLASS III
1, 2	Not Permitted	Not Permitted	Permitted
3, 4 (except Marine 4)	Not Permitted	Permitted ^c	Permitted
Marine 4, 5, 6, 7, 8	Permitted ^b	Permitted ^c	See Table R702.7(3)

a. Class I and II vapor retarders with vapor permeance greater than 1 perm when measured by ASTM E96 water method (Procedure B) shall be allowed on the interior side of any frame wall in all climate zones.

b. Use of a Class I interior vapor retarder in frame walls with a Class I vapor retarder on the exterior side shall require an approved design.

c. Where a Class II vapor retarder is used in combination with foam plastic insulating sheathing or insulated siding installed as continuous insulation on the exterior side of frame walls, the continuous insulation shall comply with Table R702.7(4) and the Class II vapor retarder shall have a vapor permeance greater than 1 perm when measured by ASTM E96 water method (Procedure B).

TABLE R702.7(3)

CLASS III VAPOR RETARDERS

CLIMATE ZONE	CLASS III VAPOR RETARDERS PERMITTED FOR: ^{a, b}
Marine 4	Vented cladding over wood structural panels.
	Vented cladding over fiberboard.
	Vented cladding over gypsum.
	Continuous insulation with <i>R</i> -value ≥ 2.5 over 2 × 4 wall.
	Continuous insulation with <i>R</i> -value ≥ 3.75 over 2 × 6 wall.
5	Vented cladding over wood structural panels.
	Vented cladding over fiberboard.
	Vented cladding over gypsum.
	Continuous insulation with <i>R</i> -value ≥ 5 over 2 × 4 wall.
	Continuous insulation with <i>R</i> -value ≥ 7.5 over 2 × 6 wall.
6	Vented cladding over fiberboard.
	Vented cladding over gypsum.
	Continuous insulation with <i>R</i> -value ≥ 7.5 over 2 × 4 wall.
	Continuous insulation with <i>R</i> -value ≥ 11.25 over 2 × 6 wall.
7	Continuous insulation with <i>R</i> -value ≥ 10 over 2 × 4 wall.
	Continuous insulation with <i>R</i> -value ≥ 15 over 2 × 6 wall.
8	Continuous insulation with <i>R</i> -value ≥ 12.5 over 2 × 4 wall.
	Continuous insulation with <i>R</i> -value ≥ 20 over 2 × 6 wall.

For SI: 1 pound per cubic foot = 16 kg/m³.

- a. Vented cladding shall include vinyl, polypropylene, or horizontal aluminum siding, brick veneer with a clear airspace as specified in Table R703.8.4(1), or other approved vented claddings.
- b. The requirements in this table apply only to insulation used to control moisture in order to permit the use of Class III vapor retarders. The insulation materials used to satisfy this option also contribute to but do not supersede the thermal envelope requirements of Chapter 11.

TABLE R702.7(4)

CONTINUOUS INSULATION WITH CLASS II VAPOR RETARDER

CLIMATE ZONE	CLASS II VAPOR RETARDERS PERMITTED FOR ^a
3	Continuous insulation with R-value ≥ 2
4, 5 and 6	Continuous insulation with R-value ≥ 3 over 2 × 4 wall. Continuous insulation with R-value ≥ 5 over 2 × 6 wall.
7	Continuous insulation with R-value ≥ 5 over 2 × 4 wall. Continuous insulation with R-value ≥ 7.5 over 2 × 6 wall.
8	Continuous insulation with R-value ≥ 7.5 over 2 × 4 wall. Continuous insulation with R-value ≥ 10 over 2 × 6 wall.

- a. The requirements in this table apply only to insulation used to control moisture in order to permit the use of Class II vapor retarders. The insulation materials used to satisfy this option also contribute to but do not supersede the thermal envelope requirements of Chapter 11.

Section R703 Exterior Covering

R703.1 General

Exterior walls shall provide the building with a weather-resistant exterior wall envelope. The exterior wall envelope shall include flashing as described in Section R703.4.

Exception: Log walls designed and constructed in accordance with the provisions of ICC 400.

R703.1.1 Water Resistance

The exterior wall envelope shall be designed and constructed in a manner that prevents the accumulation of water within the wall assembly by providing a water-resistant barrier behind the exterior cladding as required by Section R703.2 and a means of draining to the exterior water that penetrates the exterior cladding.

Exceptions:

1. A weather-resistant exterior wall envelope shall not be required over concrete or masonry walls designed in accordance with Chapter 6 and flashed in accordance with Section R703.4.
- 2.

Compliance with the requirements for a means of drainage, and the requirements of Sections R703.2 and R703.4, shall not be required for an exterior wall envelope that has been demonstrated to resist wind-driven rain through testing of the exterior wall envelope, including joints, penetrations and intersections with dissimilar materials, in accordance with ASTM E331 under the following conditions:

- 2.1. Exterior wall envelope test assemblies shall include at least one opening, one control joint, one wall/eave interface and one wall sill. All tested openings and penetrations shall be representative of the intended end-use configuration.
- 2.2. Exterior wall envelope test assemblies shall be at least 4 feet by 8 feet (1219 mm by 2438 mm) in size.
- 2.3. Exterior wall assemblies shall be tested at a minimum differential pressure of 6.24 pounds per square foot (299 Pa).
- 2.4. Exterior wall envelope assemblies shall be subjected to the minimum test exposure for a minimum of 2 hours.

The exterior wall envelope design shall be considered to resist wind-driven rain where the results of testing indicate that water did not penetrate control joints in the exterior wall envelope, joints at the perimeter of openings penetration or intersections of terminations with dissimilar materials.

R703.1.2 Wind Resistance

Wall coverings, backing materials and their attachments shall be capable of resisting wind loads in accordance with Tables R301.2(2) and R301.2(3) for walls using an effective wind area of 10 square feet. Wind-pressure resistance of the siding and backing materials shall be determined by ASTM E330 or other applicable standard test methods. Where wind-pressure resistance is determined by design analysis, data from approved design standards and analysis conforming to generally accepted engineering practice shall be used to evaluate the siding and backing material and its fastening. All applicable failure modes including bending rupture of siding, fastener withdrawal and fastener head pull-through shall be considered in the testing or design analysis. Where the wall covering and the backing material resist wind load as an assembly, use of the design capacity of the assembly shall be permitted.

R703.1.2.1 Wind Resistance of Exterior Soffits

Exterior soffits and their attachments shall comply with Section R704.

R703.2 Water-Resistive Barrier

Not fewer than one layer of *water-resistive barrier* shall be applied over studs or sheathing of all exterior walls with flashing as indicated in Section R703.4, in such a manner as to provide a continuous water-resistive barrier behind the exterior wall veneer. The water-resistive barrier material shall be continuous to the top of walls and terminated at penetrations and building appendages in a manner to meet the requirements of the exterior wall envelope as described in Section R703.1. Water-resistive barrier materials shall comply with one of the following:

1. No. 15 felt complying with ASTM D226, Type 1.
2. ASTM E2568, Type 1 or 2.
3. ASTM E331 in accordance with Section R703.11.
4. Other approved materials in accordance with the manufacturer's installation instructions.

No.15 asphalt felt and *water-resistive barriers* complying with ASTM E2556 shall be applied horizontally, with the upper layer lapped over the lower layer not less than 2 inches (51 mm), and where joints occur, shall be lapped not less than 6 inches (152 mm).

R703.3 Nominal Thickness and Attachments

The nominal thickness and attachment of exterior wall coverings shall be in accordance with Table R703.3(1), the wall covering material requirements of this section, and the wall covering manufacturer's installation instructions. Cladding attachment over foam sheathing shall comply with the additional requirements and limitations of Sections R703.15 through R703.17. Nominal material thicknesses in Table R703.3(1) are based on a maximum stud spacing of 16 inches (406 mm) on center. Where specified by the siding manufacturer's instructions and supported by a test report or other documentation, attachment to studs with greater spacing is permitted. Fasteners for exterior wall coverings attached to wood framing shall be in accordance with Section R703.3.2 and Table R703.3(1). Exterior wall coverings shall be attached to cold-formed steel light frame construction in accordance with the cladding manufacturer's installation instructions, the requirements of Table R703.3(1) using screw fasteners substituted for the nails specified in accordance with Table R703.3(2), or an approved design.

TABLE R703.3(1)

SIDING MINIMUM ATTACHMENT AND MINIMUM THICKNESS

SIDING MATERIAL		NOMINAL THICKNESS (inches)	JOINT TREATMENT	TYPE OF SUPPORTS FOR THE SIDING MATERIAL AND FASTENERS					
				Wood or wood structural panel sheathing into stud	Fiberboard sheathing into stud	Gypsum sheathing into stud	Foam plastic sheathing into stud ¹	Direct to studs	Number or spacing of fasteners
Anchored veneer: brick, concrete, masonry or stone (see Section R703.8)		2	Section R703.8	Section R703.8					
Adhered veneer: concrete, stone or masonry (see Section R703.12)		—	Section R703.12	Section R703.12					
Fiber cement siding	Panel siding (see Section R703.10.1)	⁵ / ₁₆	Section R703.10.1	6d common (2" × 0.113")	6d common (2" × 0.113")	6d common (2" × 0.113")	6d common (2" × 0.113")	4d common (1 ¹ / ₂ " × 0.099")	6" panel edges 12" inter. sup.
	Lap siding (see Section R703.10.2)	⁵ / ₁₆	Section R703.10.2	6d common (2" × 0.113")	6d common (2" × 0.113")	6d common (2" × 0.113")	6d common (2" × 0.113")	6d common (2" × 0.113") or 11 gage roofing nail	Note f
Hardboard panel siding (see Section R703.5)		⁷ / ₁₆	—	0.120" nail (shank) with 0.225" head	0.120" nail (shank) with 0.225" head	0.120" nail (shank) with 0.225" head	0.120" nail (shank) with 0.225" head	0.120" nail (shank) with 0.225" head	6" panel edges 12" inter. sup.d
Hardboard lap siding (see Section R703.5)		⁷ / ₁₆	Note e	0.099" nail (shank) with 0.240" head	0.099" nail (shank) with 0.240" head	0.099" nail (shank) with 0.240" head	0.099" nail (shank) with 0.240" head	0.099" nail (shank) with 0.240" head	Same as stud spacing 2 per bearing
Horizontal aluminum ^a	Without insulation	0.019 ^b	Lap	Siding nail 1 ¹ / ₂ " × 0.120"	Siding nail 2" × 0.120"	Siding nail 2" × 0.120"	Siding nail ^h 1 ¹ / ₂ " × 0.120"	Not allowed	Same as stud spacing
		0.024	Lap	Siding nail 1 ¹ / ₂ " × 0.120"	Siding nail 2" × 0.120"	Siding nail 2" × 0.120"	Siding nail ^h 1 ¹ / ₂ " × 0.120"	Not allowed	
	With insulation	0.019	Lap	Siding nail 1 ¹ / ₂ " × 0.120"	Siding nail 2 ¹ / ₂ " × 0.120"	Siding nail 2 ¹ / ₂ " × 0.120"	Siding nail ^h 1 ¹ / ₂ " × 0.120"	Siding nail 1 ¹ / ₂ " × 0.120"	

Insulated vinyl siding ^j		0.035 (vinyl siding layer only)	Lap	0.120 nail (shank) with a 0.313 head or 16-gage crown ^{h, i}	0.120 nail (shank) with a 0.313 head or 16-gage crown ^h	0.120 nail (shank) with a 0.313 head or 16-gage crown ^h	0.120 nail (shank) with a 0.313 head Section R703.11.2	Not allowed	16 inches on center or specified by manufacturer instructions, test report or other sections of this code
Polypropylene siding ^k		Not applicable	Lap	Section 703.14.1	Section 703.14.1	Section 703.14.1	Section 703.14.1	Not allowed	As specified by the manufacturer instructions, test report or other sections of this code
Steel ^c		29 ga.	Lap	Siding nail (1 ³ / ₄ " × 0.113")	Siding nail (2 ³ / ₄ " × 0.113")	Siding nail (2 ¹ / ₂ " × 0.113")	Siding nail (1 ³ / ₄ " × 0.113")	Not allowed	Same as stud spacing
Vinyl siding (see Section R703.11)		0.035	Lap	See Section R703.11.					
Wood siding (see Section R703.5)	Wood rustic, drop	3/ ₈ min.	Lap	6d box or siding nail (2" × 0.099")	6d box or siding nail (2" × 0.099")	6d box or siding nail (2" × 0.099")	6d box or siding nail (2" × 0.099")	8d box or siding nail (2 ¹ / ₂ " × 0.113")	Face nailing up to 6" widths, 1 nail per bearing; 8" widths and over, 2 nails per bearing
	Shiplap	¹⁹ / ₃₂ average	Lap						
	Bevel	⁷ / ₁₆							
	Butt tip	3/ ₁₆	Lap						
Wood structural panel ANSI/APA PRP 210 siding (exterior grade) (see Section R703.5)		3/ ₈ —1/ ₂	Note e	2" × 0.099" siding nail	2 ¹ / ₂ " × 0.113" siding nail	2 ¹ / ₂ " × 0.113" siding nail	2 ¹ / ₂ " × 0.113" siding nail	2" × 0.099" siding nail	6" panel edges 12" inter. sup.
Wood structural panel lap siding (see Section R703.5)		3/ ₈ —1/ ₂	Note e Note g	2" × 0.099" siding nail	2 ¹ / ₂ " × 0.113" siding nail	2 ¹ / ₂ " × 0.113" siding nail	2 ¹ / ₂ " × 0.113" siding nail	2" × 0.099" siding nail	8" along bottom edge

For SI: 1 inch = 25.4 mm.

- a. Aluminum nails shall be used to attach aluminum siding.
- b. Aluminum (0.019 inch) shall be unbacked only where the maximum panel width is 10 inches and the maximum flat area is 8 inches. The tolerance for aluminum siding shall be +0.002 inch of the nominal dimension.
- c. Shall be of approved type.
- d. Where used to resist shear forces, the spacing must be 4 inches at panel edges and 8 inches on interior supports.
- e. Vertical end joints shall occur at studs and shall be covered with a joint cover or shall be caulked.
- f. Face nailing: one 6d common nail through the overlapping planks at each stud. Concealed nailing: one 11-gage 1¹/₂ -inch-long galv. roofing nail through the top edge of each plank at each stud in accordance with the manufacturer's installation instructions.
- g. Vertical joints, if staggered, shall be permitted to be away from studs if applied over wood structural panel sheathing.
- h. Minimum fastener length must be sufficient to penetrate sheathing other nailable substrate and framing a total of a minimum of 1¹/₄ inches or in accordance with the manufacturer's installation instructions.
- i. Where specified by the manufacturer's instructions and supported by a test report, fasteners are permitted to penetrate into or fully through nailable sheathing or other nailable substrate of minimum thickness specified by the instructions or test report, without penetrating into framing.
- j. Insulated vinyl siding shall comply with ASTM D7793.
- k. Polypropylene siding shall comply with ASTM D7254.
- l. Cladding attachment over foam sheathing shall comply with the additional requirements and limitations of Sections R703.15, R703.16 and R703.17.

TABLE R703.3(2)

SCREW FASTENER SUBSTITUTION FOR SIDING ATTACHMENT TO COLD-FORMED STEEL LIGHT FRAME CONSTRUCTION^{a, b, c, d, e}

NAIL DIAMETER PER TABLE R703.3(1)	MINIMUM SCREW FASTENER SIZE
0.099"	No. 6
0.113"	No. 7
0.120"	No. 8

For SI: 1 inch = 25.4 mm

- a. Screws shall comply with ASTM C1513 and shall penetrate a minimum of three threads through minimum 33 mil (20 gage) cold-formed steel frame construction.
- b. Screw head diameter shall be not less than the nail head diameter required by Table R703.3(1).
- c. Number and spacing of screw fasteners shall comply with Table R703.3(1).
- d. Pan head, hex washer head, modified truss head or other screw head types with a flat attachment surface under the head shall be used for vinyl siding attachment.
- e. Aluminum siding shall not be fastened directly to cold-formed steel light frame construction.

R703.3.1 Wind Limitations

Where the design wind pressure exceeds 30 psf or where the limits of Table R703.3.1 are exceeded, the attachment of wall coverings shall be designed to resist the component and cladding loads specified in Table R301.2(2), adjusted for height and exposure in accordance with Table R301.2(3). For the determination of wall covering attachment, component and cladding loads shall be determined using an effective wind area of 10 square feet (0.93 m²).

TABLE R703.3.1

LIMITS FOR ATTACHMENT PER TABLE R703.3(1)

MAXIMUM MEAN ROOF HEIGHT			
Ultimate Wind Speed (mph 3-second gust)	Exposure		
	B	C	D
115	NL	50'	20'
120	NL	30'	DR
130	60'	15'	DR
140	35'	DR	DR

For SI: 1 foot = 304.8 mm, 1 mile per hour = 0.447 m/s.

NL = Not limited by Table R703.3.1, DR = Design required.

R703.3.2 Fasteners

Exterior wall coverings shall be securely fastened with aluminum, galvanized, stainless steel or rust-preventative coated nails in accordance with Table R703.3(1) or with other approved corrosion-resistant fasteners in accordance with the wall covering manufacturer's installation instructions. Nails shall comply with ASTM F1667. Nails shall be T-head, modified round head, or round head with smooth or deformed shanks. Where fiberboard, gypsum, or foam plastic sheathing backing is used, nails shall be driven into the studs. Where wood or wood structural panel sheathing is used, fasteners shall be driven into studs unless otherwise permitted to be driven into sheathing in accordance with either the siding manufacturer's installation instructions or Table R703.3.2.

TABLE R703.3.2

OPTIONAL SIDING ATTACHMENT SCHEDULE FOR FASTENERS WHERE NO STUD PENETRATION NECESSARY

APPLICATION	NUMBER AND TYPE OF FASTENER	SPACING OF FASTENERS ^b
Exterior wall covering (weighing 3 psf or less) attachment to wood structural panel sheathing, either direct or over foam sheathing a maximum of 2 inches thick. ^a Note: Does not apply to vertical siding.	Ring shank roofing nail (0.120" min. dia.)	12" o.c.
	Ring shank nail (0.148" min. dia.)	15" o.c.
	No. 6 screw (0.138" min. dia.)	12" o.c.
	No. 8 screw (0.164" min. dia.)	16" o.c.

For SI: 1 inch = 25.4 mm.

- a. Fastener length shall be sufficient to penetrate back side of the wood structural panel sheathing by at least $\frac{1}{4}$ inch. The wood structural panel sheathing shall be not less than $\frac{7}{16}$ inch in thickness.
- b. Spacing of fasteners is per 12 inches of siding width. For other siding widths, multiply "Spacing of Fasteners" above by a factor of 12/s, where "s" is the siding width in inches. Fastener spacing shall never be greater than the manufacturer's minimum recommendations.

R703.3.3 Minimum Fastener Length and Penetration

Fasteners shall have the greater of the minimum length specified in Table R703.3(1) or as required to provide a minimum penetration into framing as follows:

- Fasteners for horizontal aluminum siding, steel siding, particleboard panel siding, wood structural panel siding in accordance with ANSI/APA-PRP 210, fiber-cement panel siding and fiber-cement lap siding installed over foam plastic sheathing shall penetrate not less than $1\frac{1}{2}$ inches (38 mm) into framing or shall be in accordance with the manufacturer's installation instructions.
- Fasteners for hardboard panel and lap siding shall penetrate not less than $1\frac{1}{2}$ inches (38 mm) into framing.
- Fasteners for vinyl siding and insulated vinyl siding shall be installed in accordance with Section R703.11 or R703.13.

4. Fasteners for vertical or horizontal wood siding shall penetrate not less than $\frac{1}{2}$ inches (38 mm) into studs, studs and wood sheathing combined, or blocking.
5. Fasteners for siding material installed over foam plastic sheathing shall have sufficient length to accommodate foam plastic sheathing thickness and to penetrate framing or sheathing and framing combined, as specified in Items 1 through 4.

R703.3.4

Fasteners for polypropylene siding shall be installed in accordance with Section R703.14.

R703.3.5 Siding Clearance at Wall and Adjacent Surfaces

Unless otherwise specified by the material manufacturer, or this code, siding shall have a clearance of at least 6 inches (152 mm) from grade and at least $\frac{1}{2}$ inch (13 mm) from other adjacent surfaces (decks, roofs, slabs).

R703.4 Flashing

Approved metal flashing, vinyl flashing, self-adhered membranes and mechanically attached flexible flashing shall be applied shingle-fashion or in accordance with the manufacturer's instructions. Metal flashing shall be corrosion resistant. Fluid-applied membranes used as flashing shall be applied in accordance with the manufacturer's instructions. All flashing shall be applied in a manner to prevent the entry of water into the wall cavity or penetration of water to the building structural framing components. Self-adhered membranes used as flashing shall comply with AAMA 711. All exterior fenestration products shall be sealed at the juncture with the building wall with a sealant complying with AAMA 800 or ASTM C920 Class 25 Grade NS or greater for proper joint expansion and contraction, ASTM C1281, AAMA 812, or other approved standard as appropriate for the type of sealant. Fluid-applied membranes used as flashing in exterior walls shall comply with AAMA 714. The flashing shall extend to the surface of the exterior wall finish. Approved flashings shall be installed at the following locations:

1. Exterior window and door openings. Flashing at exterior window and door openings shall extend to the surface of the exterior wall finish or to the water-resistive barrier complying with Section 703.2 for subsequent drainage. Mechanically attached flexible flashings shall comply with AAMA 712. Flashing at exterior window and door openings shall be installed in accordance with one or more of the following:
 - 1.1. The fenestration manufacturer's installation and flashing instructions, or for applications not addressed in the fenestration manufacturer's instructions, in accordance with the flashing or water-resistive barrier manufacturer's instructions. Where flashing instructions or details are not provided, pan flashing shall be installed at the sill of exterior window and door openings. Pan flashing shall be sealed or sloped in such a manner as to direct water to the surface of the exterior wall finish or to the water-resistive barrier for subsequent drainage. Openings using pan flashing shall incorporate flashing or protection at the head and sides.
 - 1.2. In accordance with the flashing design or method of a registered design professional.
 - 1.3. In accordance with other approved methods.
 - 1.4. In accordance with FMA/AAMA 100, FMA/AAMA 200, FMA/WDMA 250, FMA/AAMA/WDMA 300 or FMA/AAMA/WDMA 400, or FMA/AAMA/WDMA 2710.
2. At the intersection of chimneys or other masonry construction with frame or stucco walls, with projecting lips on both sides under stucco copings.
3. Under and at the ends of masonry, wood or metal copings and sills.
4. Continuously above all projecting wood trim.
5. Where exterior porches, decks or stairs attach to a wall or floor assembly of wood-frame construction.
6. At wall and roof intersections.
7. At built-in gutters.

R703.5 Wood, Hardboard and Wood Structural Panel Siding

Wood, hardboard, and wood structural panel siding shall be installed in accordance with this section and Table R703.3(1). Hardboard siding shall comply with CPA/ANSI A135.6.

R703.5.1 Vertical Wood Siding

Wood siding applied vertically shall be nailed to horizontal nailing strips or blocking set not more than 24 inches (610 mm) on center.

R703.5.2 Panel Siding

$\frac{3}{8}$ -inch (9.5 mm) wood structural panel siding shall not be applied directly to studs spaced more than 16 inches (406 mm) on center where long dimension is parallel to studs. Wood structural panel siding $\frac{7}{16}$ inch (11.1 mm) or thinner shall not be applied directly to studs spaced more than 24 inches (610 mm) on center. The stud spacing shall not exceed the panel span rating provided by the manufacturer unless the panels are installed with the face grain perpendicular to the studs or over sheathing approved for that stud spacing.

Joints in wood, hardboard or wood structural panel siding shall be made as follows unless otherwise approved. Vertical joints in panel siding shall occur over framing members, unless wood or wood structural panel sheathing is used, and shall be shiplapped or covered with a batten. Horizontal joints in panel siding shall be lapped not less than 1 inch (25 mm) or shall be shiplapped or flashed with Z-flashing and occur over solid blocking, wood or wood structural panel sheathing.

R703.5.3 Horizontal Wood Siding

Horizontal lap siding shall be installed in accordance with the manufacturer's recommendations. Where there are no recommendations the siding shall be lapped not less than 1 inch (25 mm), or $\frac{1}{2}$ inch (12.7 mm) if rabbeted, and shall have the ends caulked, covered with a batten or sealed and installed over a strip of flashing.

R703.6 Wood Shakes and Shingles

Wood shakes and shingles shall conform to CSSB *Grading Rules for Wood Shakes and Shingles*.

R703.6.1 Application

Wood shakes or shingles shall be applied either single course or double course over nominal $\frac{1}{2}$ -inch (12.7 mm) wood-based sheathing or to furring strips over $\frac{1}{2}$ -inch (12.7 mm) nominal nonwood sheathing. A water-resistive barrier shall be provided over all sheathing, with horizontal overlaps in the membrane of not less than 2 inches (51 mm) and vertical overlaps of not less than 6 inches (152 mm). Where horizontal furring strips are used, they shall be 1 inch by 3 inches or 1 inch by 4 inches (25 mm by 76 mm or 25 mm by 102 mm) and shall be fastened to the studs with minimum 7d or 8d box nails and shall be spaced a distance on center equal to the actual weather exposure of the shakes or shingles, not to exceed the maximum exposure specified in Table R703.6.1. When installing shakes or shingles over a nonpermeable water-resistive barrier, furring strips shall be placed first

vertically over the barrier and in addition, horizontal furring strips shall be fastened to the vertical furring strips prior to attaching the shakes or shingles to the horizontal furring strips. The spacing between adjacent shingles to allow for expansion shall be $\frac{1}{8}$ inch (3.2 mm) to $\frac{1}{4}$ inch (6.4 mm) apart, and between adjacent shakes shall be $\frac{3}{8}$ inch (9.5 mm) to $\frac{1}{2}$ inch (12.7 mm) apart. The offset spacing between joints in adjacent courses shall be not less than $\frac{1}{2}$ inches (38 mm).

TABLE R703.6.1

MAXIMUM WEATHER EXPOSURE FOR WOOD SHAKES AND SHINGLES ON EXTERIOR WALLS^{a, b, c}

(Dimensions are in inches)

LENGTH	EXPOSURE FOR SINGLE COURSE	EXPOSURE FOR DOUBLE COURSE
Shingles ^a		
16	7	12 ^b
18	8	14 ^c
24	10 $\frac{1}{2}$	16 ^d
Shakes ^a		
18	8	14
24	10 $\frac{1}{2}$	18

For SI: 1 inch = 25.4 mm.

- a. Dimensions given are for No. 1 grade.
- b. A maximum 9-inch exposure is permitted for No. 2 grade.
- c. A maximum 10-inch exposure is permitted for No. 2 grade.
- d. A maximum 14-inch exposure is permitted for No. 2 grade.

R703.6.2 Weather Exposure

The maximum weather exposure for shakes and shingles shall not exceed that specified in Table 703.6.1.

R703.6.3 Attachment

Wood shakes or shingles shall be installed according to this chapter and the manufacturer's instructions. Where wind pressures determined in accordance with Table R301.2(2) do not exceed 30 psf, each shake or shingle shall be held in place by two stainless steel Type 304, Type 316 or hot-dipped zinc-coated galvanized corrosion-resistant box nails in accordance with Table R703.6.3(1) or R703.6.3(2). The hot-dipped zinc-coated galvanizing shall conform to minimum standard ASTM A153D, 1.0 ounce per square foot. In single-course application, the fasteners shall be concealed by the course above and shall be driven approximately 1 inch (25 mm) above the butt line of the succeeding course and $\frac{3}{4}$ inch (19 mm) from the edge. In double-course applications, the exposed shake or shingle shall be face-nailed with two fasteners, driven approximately 2 inches (51 mm) above the butt line and $\frac{3}{4}$ inch (19 mm) from each edge. Fasteners installed within 15 miles (24 km) of salt water coastal areas shall be stainless steel Type 316. Fasteners for fire-retardant-treated shakes or shingles in accordance with Section R902 or pressure-impregnated-preservative-treated shakes or shingles in accordance with AWPA U1 shall be stainless steel Type 316. The fasteners shall penetrate the sheathing or furring strips by not less than $\frac{1}{2}$ inch (13 mm) and shall not be overdriven. Fasteners for untreated (natural) and treated products shall comply with ASTM F1667.

TABLE R703.6.3(1)

SINGLE COURSE SIDEWALL FASTENERS

PRODUCT TYPE	NAIL TYPE AND MINIMUM LENGTH (inches)	MINIMUM HEAD DIAMETER (inches)	MINIMUM SHANK THICKNESS (inches)
R & R and sanded shingles	Type		
16" and 18" shingles	3d box $1\frac{1}{4}$	0.19	0.08
24" shingles	4d box $1\frac{1}{2}$	0.19	0.08
Grooved shingles	Type		
16" and 18" shingles	3d box $1\frac{1}{4}$	0.19	0.08
24" shingles	4d box $1\frac{1}{2}$	0.19	0.08
Split and sawn shakes	Type		
18" straight-split shakes	5d box $1\frac{3}{4}$	0.19	0.08
18" and 24" handsplit shakes	6d box 2	0.19	0.0915
24" tapersplit shakes	5d box $1\frac{3}{4}$	0.19	0.08
18" and 24" tapersawn shakes	6d box 2	0.19	0.0915

For SI: 1 inch = 25.4 mm.

TABLE R703.6.3(2)

DOUBLE COURSE SIDEWALL FASTENERS

PRODUCT TYPE	NAIL TYPE AND MINIMUM LENGTH	MINIMUM HEAD DIAMETER (inches)	MINIMUM SHANK THICKNESS (inches)
R & R and sanded shingles			
16," 8" and 24" shingles	5d box 1 ³ / ₄ or same size casing nails	0.19	0.08
Grooved shingles			
16," 18" and 24" shingles	5d box 1 ³ / ₄	0.19	0.08
Split and sawn shakes			
18" straight-split shakes	7d box 2 ¹ / ₄ or 8d 2 ¹ / ₂	0.19	0.099
18" and 24" handsplit shakes	7d box 2 ¹ / ₄ or 8d 2 ¹ / ₂	0.19	0.099
24" tapersplit shakes	7d box 2 ¹ / ₄ or 8d 2 ¹ / ₂	0.19	0.099
18" and 24" tapersawn shakes	7d box 2 ¹ / ₄ or 8d 2 ¹ / ₂	0.19	0.099

For SI: 1 inch = 25.4 mm.

R703.6.4 Bottom Courses

The bottom courses shall be doubled.

R703.7 Exterior Plaster

Installation of these materials shall be in compliance with ASTM C926, ASTM C1063 or ASTM C1787 and the provisions of this code.

R703.7.1 Lath

Lath and lath attachments shall be of corrosion-resistant materials. Expanded metal or woven wire lath shall be attached with 1¹/₂-inch-long (38 mm), 11 gage nails having a ⁷/₁₆-inch (11.1 mm) head, or 1¹/₂-inch-long (22.2 mm), 16 gage staples, spaced in accordance with ASTM C1063 or C1787, or as otherwise approved.

R703.7.2 Plaster

Plastering with cement plaster shall be not less than three coats where applied over any type of code-approved lath and shall be not less than two coats where directly applied over masonry, concrete, clay brick, stone or tile. If the plaster surface is completely covered by veneer or other facing material or is completely concealed, plaster application need be only two coats, provided the total thickness is as set forth in Table R702.1(1).

On wood-frame construction with an on-grade floor slab system, exterior plaster shall be applied to cover, but not extend below, lath, paper and screed. Cement plaster shall be in accordance with ASTM C926. Cement materials shall be in accordance with one of the following:

1. Masonry cement conforming to ASTM C91 Type M, S or N.
2. Portland cement conforming to ASTM C150 Type I, II or III.
3. Blended hydraulic cement conforming to ASTM C595 Type IP, IS(S<70), IL or IT(S<70).
4. Hydraulic cement conforming to ASTM C1157 Type GU, HE, MS, HS or MH.
5. Plaster (stucco) cement conforming to ASTM C1328.

The proportion of aggregate to cementitious materials shall be as set forth in Table R702.1(3).

R703.7.2.1 Weep Screeds

A minimum 0.019-inch (0.5 mm) (No. 26 galvanized sheet gage), corrosion-resistant weep screed or plastic weep screed, with a minimum vertical attachment flange of 3¹/₂ inches (89 mm) shall be provided at or below the foundation plate line on exterior stud walls in accordance with ASTM C926. The weep screed shall be placed not less than 4 inches (102 mm) above the earth or 2 inches (51 mm) above paved areas and shall be of a type that will allow trapped water to drain to the exterior of the building. The weather-resistant barrier shall lap the attachment flange. The exterior lath shall cover and terminate on the attachment flange of the weep screed.

R703.7.3 Water-Resistive Barriers

Water-resistive barriers shall be installed as required in Section R703.2 and, where applied over wood-based sheathing, shall include a water-resistive vapor-permeable barrier with a performance at least equivalent to two layers of Grade D paper. The individual layers shall be installed independently such that each layer provides a separate continuous plane and any flashing (installed in accordance with Section R703.4) intended to drain to the water-resistive barrier is directed between the layers.

Exception: Where the water-resistive barrier that is applied over wood-based sheathing has a water resistance equal to or greater than that of 60-minute Grade D paper and is separated from the stucco by an intervening, substantially nonwater-absorbing layer or designed drainage space.

R703.7.4 Application

Each coat shall be kept in a moist condition for at least 48 hours prior to application of the next coat.

Exception: Applications installed in accordance with ASTM C926 including the reference in ASTM C926 Section 8 to Section X14.2 of the Appendix.

R703.7.5 Curing

The finish coat for two-coat cement plaster shall not be applied sooner than seven days after application of the first coat. For three-coat cement plaster, the second coat shall not be applied sooner than 48 hours after application of the first coat. The finish coat for three-coat cement plaster shall not be applied sooner than seven days after application of the second coat.

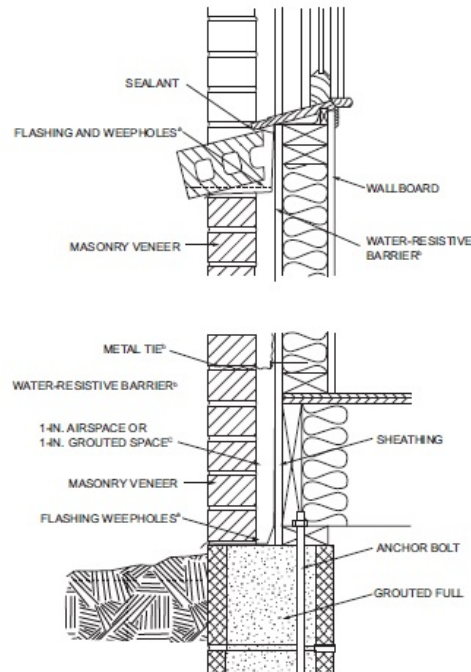
Exception: Applications installed in accordance with ASTM C926 including the reference in ASTM C926 Section 8 to Section X1.4.2 of the Appendix.

R703.8 Anchored Stone and Masonry Veneer, General

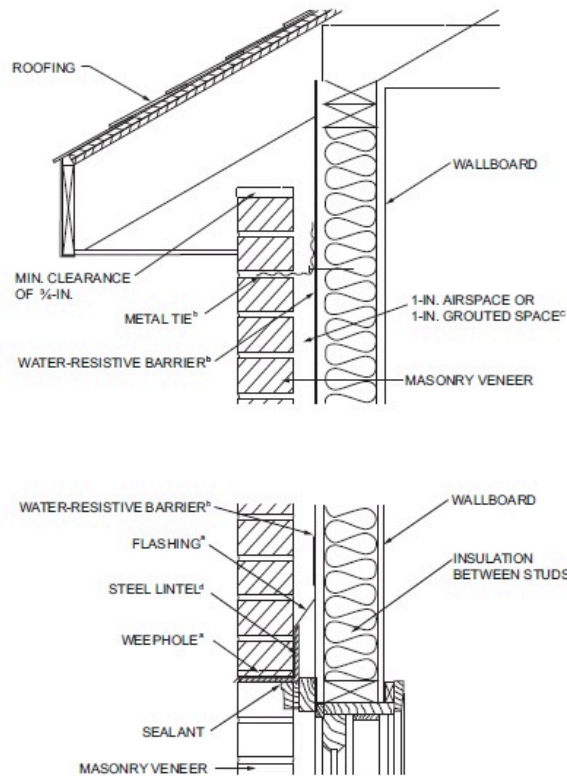
Anchored stone and masonry veneer shall be installed in accordance with this chapter, Table R703.3(1) and Figure R703.8. These veneers installed over a backing of wood or cold-formed steel shall be limited to the first story above grade plane and shall not exceed 5 inches (127 mm) in thickness. See Section R602.3 for wall bracing requirements for masonry veneer for wood-framed construction and Section R603.1 for wall bracing requirements for masonry veneer for cold-formed steel construction. The provisions of this section are limited to areas where the ultimate design wind speed, V_{ult} , is less than 165 mph. Where the ultimate design wind speed, V_{ult} , equals or exceeds 165 mph, anchored stone and masonry veneer shall comply with TMS 402/ACI 530/ASCE 5.

Exceptions:

1. Exterior stone or masonry veneer, as specified in Table R703.8(1), with a backing of wood or steel framing shall be permitted to the height specified in Table R703.8(1) above a noncombustible foundation.
2. Reserved.



For SI: 1 inch = 24.5 mm.



For SI: 1 inch = 25.4 mm.

- a. See Sections R703.8.5, R703.8.6 and R703.4.
- b. See Sections R703.2 and R703.8.4.
- c. See Section R703.8.4.2 and Table R703.8.4(1).
- d. See Section R703.8.3.
- e. Figure R703.8 illustrates typical construction details for a masonry veneer wall. For the actual mandatory requirements of this code, see the indicated sections of text. Other details of masonry veneer wall construction shall be permitted provided the requirements of the indicated sections of text are met.

FIGURE R703.8

TYPICAL MASONRY VENEER WALL DETAILS^e

TABLE R703.8(1)

STONE OR MASONRY VENEER LIMITATIONS AND REQUIREMENTS, WOOD OR STEEL FRAMING

NUMBER OF WOOD- OR STEEL-FRAMED STORIES	MAXIMUM HEIGHT OF VENEER ABOVE NONCOMBUSTIBLE FOUNDATION ^a (feet)	MAXIMUM NOMINAL THICKNESS OF VENEER (inches)	MAXIMUM WEIGHT OF VENEER (psf) ^b	WOOD- OR STEEL-FRAMED STORY
Steel: 1 or 2 Wood: 1, 2 or 3	30	5	50	all

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm, 1 pound per square foot = 0.479 kPa.

- a. An additional 8 feet is permitted for gable end walls. See also story height limitations of Section R301.3.
- b. Maximum weight is installed weight and includes weight of mortar, grout, lath and other materials used for installation. Where veneer is placed on both faces of a wall, the combined weight shall not exceed that specified in this table.

TABLE R703.8(2) STONE OR MASONRY VENEER LIMITATIONS AND REQUIREMENTS, ONE- AND TWO-FAMILY DETACHED DWELLINGS, SEISMIC DESIGN CATEGORIES D₀, D₁ AND D₂.

Reserved.

R703.8.1 Interior Veneer Support

Veneers used as interior wall finishes shall be permitted to be supported on wood or cold-formed steel floors that are designed to support the loads imposed.

R703.8.2 Exterior Veneer Support

Exterior masonry veneers having an installed weight of 40 pounds per square foot (195 kg/m²) or less shall be permitted to be supported on wood or cold-formed steel construction. Where masonry veneer supported by wood or cold-formed steel construction adjoins masonry veneer supported by the foundation, there shall be a movement joint between the veneer supported by the wood or cold-formed steel construction and the veneer supported by the foundation. The wood or cold-formed steel construction supporting the masonry veneer shall be designed to limit the deflection to $\frac{1}{600}$ of the span for the supporting members. The design of the wood or cold-formed steel construction shall consider the weight of the veneer and any other loads.

R703.8.2.1 Support by Steel Angle

A minimum 6-inch by 4-inch by $\frac{5}{16}$ -inch (152 mm by 102 mm by 8 mm) steel angle, with the long leg placed vertically, shall be anchored to double 2-inch by 4-inch (51 mm by 102 mm) wood studs or double 350S162 cold-formed steel studs at a maximum on-center spacing of 16 inches (406 mm). Anchorage of the steel angle at every double stud spacing shall be not less than two $\frac{7}{16}$ -inch-diameter (11 mm) by 4-inch (102 mm) lag screws for wood construction or two $\frac{7}{16}$ -inch (11.1 mm) bolts with washers for cold-formed steel construction. The steel angle shall have a minimum clearance to underlying construction of $\frac{1}{16}$ inch (1.6 mm). Not less than two-thirds the width of the masonry veneer thickness shall bear on the steel angle. Flashing and weep holes shall be located in the masonry veneer in accordance with Figure R703.8.2.1. The maximum height of masonry veneer above the steel angle support shall be 12 feet 8 inches (3861 mm). The airspace separating the masonry veneer from the wood backing shall be in accordance with Sections R703.8.4 and R703.8.4.2. The method of support for the masonry veneer on wood construction shall be constructed in accordance with Figure R703.8.2.1.

The maximum slope of the roof construction without stops shall be 7:12. Roof construction with slopes greater than 7:12 but not more than 12:12 shall have stops of a minimum 3-inch by 3-inch by $\frac{1}{4}$ -inch (76 mm by 76 mm by 6.4 mm) steel plate welded to the angle at 24 inches (610 mm) on center along the angle or as *approved by the building official*.

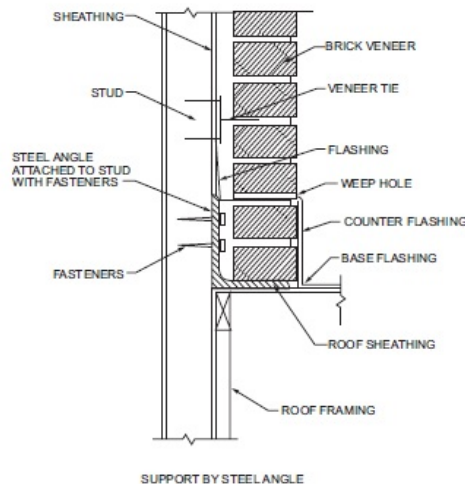


FIGURE R703.8.2.1

EXTERIOR MASONRY VENEER SUPPORT BY STEEL ANGLES

R703.8.2.2 Support by Roof Construction

A steel angle shall be placed directly on top of the roof construction. The roof supporting construction for the steel angle shall consist of not fewer than three 2-inch by 6-inch (51 mm by 152 mm) wood members for wood construction or three 550S162 cold-formed steel members for cold-formed steel light frame construction. A wood member abutting the vertical wall stud construction shall be anchored with not fewer than three $\frac{5}{8}$ -inch (15.9 mm) diameter by 5-inch (127 mm) lag screws to every wood stud spacing. Each additional wood roof member shall be anchored by the use of two 10d nails at every wood stud spacing. A cold-formed steel member abutting the vertical wall stud shall be anchored with not fewer than nine No. 8 screws to every cold-formed steel stud. Each additional cold-formed steel roof member shall be anchored to the adjoining roof member using two No. 8 screws at every stud spacing. Not less than two-thirds the width of the masonry veneer thickness shall bear on the steel angle. Flashing and weep holes shall be located in the masonry veneer wythe in accordance with Figure R703.8.2.2. The maximum height of the masonry veneer above the steel angle support shall be 12 feet 8 inches (38.61 mm). The airspace separating the masonry veneer from the wood backing shall be in accordance with Sections R703.8.4 and R703.8.4.2. The support for the masonry veneer shall be constructed in accordance with Figure R703.8.2.2.

The maximum slope of the roof construction without stops shall be 7:12. Roof construction with slopes greater than 7:12 but not more than 12:12 shall have stops of a minimum 3-inch by 3-inch by $\frac{1}{4}$ -inch (76 mm by 76 mm by 6.4 mm) steel plate welded to the angle at 24 inches (610 mm) on center along the angle or as *approved by the building official*.

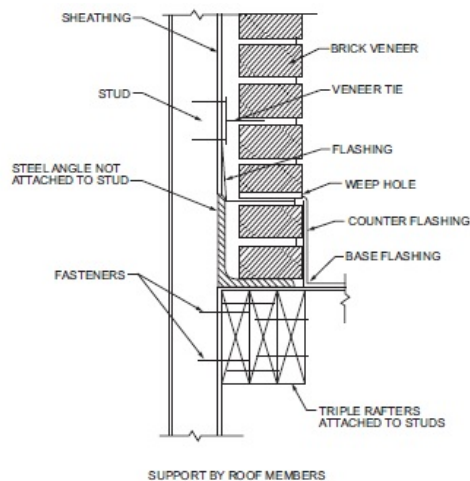


FIGURE R703.8.2.2

EXTERIOR MASONRY VENEER SUPPORT BY ROOF MEMBERS

R703.8.3 Lintels

Masonry veneer shall not support any vertical load other than the dead load of the veneer above. Veneer above openings shall be supported on lintels of non-combustible materials. The lintels shall have a length of bearing not less than 4 inches (102 mm). Steel lintels shall be shop coated with a rust-inhibitive paint, except for lintels made of corrosion-resistant steel or steel treated with coatings to provide corrosion resistance. Construction of openings shall comply with either Section R703.8.3.1 or 703.8.3.2.

R703.8.3.1 Allowable Span

The allowable span shall not exceed the values set forth in Table R703.8.3.1.

TABLE R703.8.3.1

ALLOWABLE SPANS FOR LINTELS SUPPORTING MASONRY VENEER^{a, b, c, d}

SIZE OF STEEL ANGLE ^{a, c, d} (inches)	NO STORY ABOVE	ONE STORY ABOVE	TWO STORIES ABOVE	NO. OF 1/2-INCH OR EQUIVALENT REINFORCING BARS IN REINFORCED LINTEL ^{b, d}
3 × 3 × 1/4	6'-0"	4'-6"	3'-0"	1
4 × 3 × 1/4	8'-0"	6'-0"	4'-6"	1
5 × 3 1/2 × 5/16	10'-0"	8'-0"	6'-0"	2
6 × 3 1/2 × 5/16	14'-0"	9'-6"	7'-0"	2
2-6 × 3 1/2 × 5/16	20'-0"	12'-0"	9'-6"	4

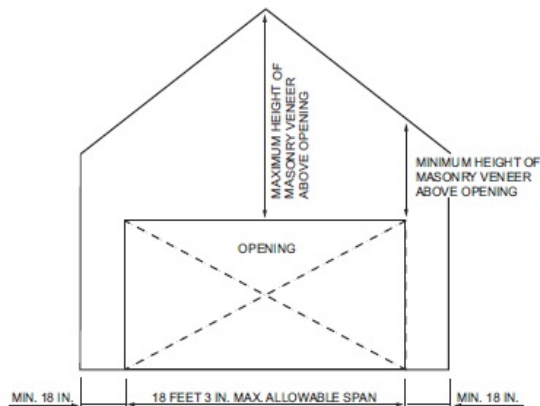
For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.

- Long leg of the angle shall be placed in a vertical position.
- Depth of reinforced lintels shall be not less than 8 inches and all cells of hollow masonry lintels shall be grouted solid. Reinforcing bars shall extend not less than 8 inches into the support.
- Steel members indicated are adequate typical examples; other steel members meeting structural design requirements shall be permitted to be used.
- Either steel angle or reinforced lintel shall span opening.

R703.8.3.2 Maximum Span

The allowable span shall not exceed 18 feet 3 inches (5562 mm) and shall be constructed to comply with Figure R703.8.3.2 and the following:

- Provide a minimum length of 18 inches (457 mm) of masonry veneer on each side of opening as shown in Figure R703.8.3.2.
- Provide a minimum 5-inch by 3 1/2-inch by 5/16-inch (127 mm by 89 mm by 7.9 mm) steel angle above the opening and shore for a minimum of 7 days after installation.
- Provide double-wire joint reinforcement extending 12 inches (305 mm) beyond each side of the opening. Lap splices of joint reinforcement not less than 12 inches (305 mm). Comply with one of the following:
 - Double-wire joint reinforcement shall be 3/16-inch (4.8 mm) diameter and shall be placed in the first two bed joints above the opening.
 - Double-wire joint reinforcement shall be 9 gauge (0.144 inch or 3.66 mm diameter) and shall be placed in the first three bed joints above the opening.
- Provide the height of masonry veneer above opening, in accordance with Table R703.8.3.2.



For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.

FIGURE R703.8.3.2

MASONRY VENEER OPENING

TABLE R703.8.3.2

HEIGHT OF MASONRY VENEER ABOVE OPENING

MINIMUM HEIGHT OF MASONRY VENEER ABOVE OPENING (INCH)	MAXIMUM HEIGHT OF MASONRY VENEER ABOVE OPENING (FEET)
13	< 5
24	5 to < 12
60	12 to height above support allowed by Section R703.8

For SI: 1 inch = 25.4 mm, 1 foot = 304.8 mm.

R703.8.4 Anchorage

Masonry veneer shall be anchored directly to the supporting wall studs with corrosion-resistant metal ties embedded in mortar or grout and extending into the veneer a minimum of 1½ inches (38 mm), with not less than ⅝-inch (15.9 mm) mortar or grout cover to outside face. Masonry veneer shall conform to Table R703.8.4(1). Where the masonry veneer tie attachment is fastened to wood structural panel not less than 7/16 performance category through insulating sheathing not greater than 2 inches (51 mm) in thickness, see Table R703.8.4(2). Where Table R703.8.4(2) is used, attachment to the wood studs behind the sheathing is not required.

TABLE R703.8.4(1)

TIE ATTACHMENT AND AIRSPACE REQUIREMENTS

BACKING AND TIE	MINIMUM TIE	MINIMUM TIE FASTENER ^a	AIRSPACE ^b	
Wood stud backing with corrugated sheet metal	22 US gage (0.0299 in.) × 7/8 in. wide	8d common nail (2½ in. × 0.131 in.)	Nominal 1 in. between sheathing and veneer	
Wood stud backing with adjustable metal strand wire	W1.7 (No. 9 US gage; 0.148 in. dia.) with hook embedded in mortar joint ^d	8d common nail (2½ in. × 0.131 in.)	Minimum nominal 1 in. between sheathing and veneer	Maximum 4⅝ in. between backing and veneer
Wood stud backing with adjustable metal strand wire	W2.8 (0.187 in. dia.) with hook embedded in mortar joint ^{e, f}	8d common nail (2½ in. × 0.131 in.)	Greater than 4⅝ in. between backing and veneer	Maximum 6⅝ in. between backing and veneer
Cold-formed steel stud backing with adjustable metal strand wire	W1.7 (No. 9 US gage; 0.148 in. dia.) with hook embedded in mortar joint ^d	No. 10 screw extending through the steel framing a minimum of three exposed threads	Minimum nominal 1 in. between sheathing and veneer	Maximum 4⅝ in. between backing and veneer
Cold-formed steel stud backing with adjustable metal strand wire	W2.8 (0.148 in. dia.) with hook embedded in mortar joint ^{e, f}	No. 10 screw extending through the steel framing a minimum of three exposed threads	Greater than 4⅝ in. between backing and veneer	Maximum 6⅝ in. between backing and veneer

For SI: 1 inch = 25.4 mm.

- a. All fasteners shall have rust-inhibitive coating suitable for the installation in which they are being used, or be manufactured from material not susceptible to corrosion.
- b. An airspace that provides drainage shall be permitted to contain some mortar from construction.
- c. Reserved.
- d. Adjustable tie pintle shall include a minimum of 1 pintle leg of wire size W2.8 (MW18) with a maximum offset of 1¼ in.
- e. Adjustable tie pintle shall include a minimum of 2 pintle legs with a maximum offset of 1¼ in. Distance between inside face of brick and end of pintle shall be a maximum of 2 in.
- f. Adjustable tie backing attachment components shall consist of one of the following: eyes with minimum wire W2.8 (MW18), barrel with minimum ¼ in. outside dia., or plate with minimum thickness of 0.074 in. and minimum width of 1¼ in.

TABLE R703.8.4(2)

REQUIRED BRICK TIE SPACING FOR DIRECT APPLICATION TO WOOD STRUCTURAL PANEL SHEATHING^{a, b, c}

FASTENER TYPE ^d	SIZE	REQUIRED BRICK-TIE SPACING (VERTICAL-TIE SPACING/HORIZONTAL-TIE SPACING) (inches/inches)
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	(DIA. OR SCREW #)	110 mph V Ultimate			115 mph V Ultimate			130 mph V Ultimate			140 mph V Ultimate		
		Zone 5, Exposure B	Zone 5, Exposure C	Zone 5, Exposure D	Zone 5, Exposure B	Zone 5, Exposure C	Zone 5, Exposure D	Zone 5, Exposure B	Zone 5, Exposure C	Zone 5, Exposure D	Zone 5, Exposure B	Zone 5, Exposure C	Zone 5, Exposure D
Ring Shank Nails	0.091	16/16, 16/12, 12/16, 12/12	16/12, 12/16, 12/12	12/12	16/16, 16/12, 12/16, 12/12	16/12, 12/16, 12/12	12/12	16/12, 12/16, 12/12	12/12	—	12/12	—	—
	0.148	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	16/12, 12/16, 12/12	16/12, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	16/12, 12/16, 12/12	12/12
Screws	#6	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	16/12, 12/16, 12/12	16/12, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	16/12, 12/16, 12/12	12/12
	#8	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	16/12, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	16/12, 12/16, 12/12	16/12, 12/16, 12/12
	#10	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	16/12, 12/16, 12/12
	#14	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	24/16, 16/24, 16/16, 16/12, 12/16, 12/16, 12/12	16/16, 16/12, 12/16, 12/12	16/16, 16/12, 12/16, 12/12

For SI: 1 inch = 25.4 mm, 1 mph = 0.447 m/s.

- This table is based on attachment of brick ties directly to wood structural panel sheathing only. Additional attachment of the brick tie to lumber framing is not required. The brick ties shall be permitted to be placed over any insulating sheathing, not to exceed 2 inches in thickness. Wood structural panel sheathing shall be a minimum 7/16 performance category. The table is based on a building height of 30 feet or less.
- Wood structural panels shall have a specific gravity of 0.42 or greater in accordance with NDS.
- Foam sheathing shall have a minimum compressive strength of 15 psi in accordance with ASTM C578 or ASTM C1289.
- Fasteners shall be sized such that the tip of the fastener passes completely through the wood structural panel sheathing by not less than $\frac{1}{4}$ inch.

R703.8.4.1 Size and Spacing

Veneer ties, if strand wire, shall be not less in thickness than No. 9 US gage [(0.148 inch) (4 mm)] wire and shall have a hook embedded in the mortar joint, or if sheet metal, shall be not less than No. 22 US gage by [(0.0299 inch) (0.76 mm)] $\frac{7}{8}$ inch (22 mm) corrugated. Each tie shall support not more than 2.67 square feet (0.25 m²) of wall area and shall be spaced not more than 32 inches (813 mm) on center horizontally and 24 inches (635 mm) on center vertically.

Exceptions:

- Reserved.
- Where the ultimate design wind speed, V_{ult} , exceeds 140 mph, each tie shall support not more than 1.8 square feet (0.167 m²) of wall area and anchors shall be spaced at a maximum 18 inches (457 mm) horizontally and vertically.

R703.8.4.1.1 Veneer Ties Around Wall Openings

Additional metal ties shall be provided around wall openings greater than 16 inches (406 mm) in either dimension. Metal ties around the perimeter of openings shall be spaced not more than 3 feet (914 mm) on center and placed within 12 inches (305 mm) of the wall opening.

R703.8.4.2 Grout Fill

As an alternative to the airspace required by Table R703.8.4(1), grout shall be permitted to fill the airspace. Where the airspace is filled with grout, a water-resistive barrier is required over studs or sheathing. Where the airspace is filled, replacing the sheathing and water-resistive barrier with a wire mesh and approved water-resistive barrier or an

approved water-resistive barrier-backed reinforcement attached directly to the studs is permitted.

R703.8.5 Flashing

Flashing shall be located beneath the first course of masonry above finished ground level above the foundation wall or slab and at other points of support, including structural floors, shelf angles and lintels where masonry veneers are designed in accordance with Section R703.8. See Section R703.4 for additional requirements.

R703.8.6 Weepholes

Weepholes shall be provided in the outside wythe of masonry walls at a maximum spacing of 33 inches (838 mm) on center. Weepholes shall be not less than $\frac{3}{16}$ inch (5 mm) in diameter. Weepholes shall be located immediately above the flashing.

R703.9 Exterior Insulation and Finish System (EIFS)/EIFS With Drainage

Exterior insulation and finish systems (EIFS) shall be designed or tested to meet the wind pressures specified in Table R301.2(2) and shall comply with this chapter and Section R703.9.1. EIFS with drainage shall comply with this chapter and Section R703.9.2.

R703.9.1 Exterior Insulation and Finish Systems (EIFS)

EIFS shall comply with the following:

1. ASTM E2568.
2. EIFS shall be limited to applications over substrates of concrete or masonry wall assemblies.
3. Flashing of EIFS shall be provided in accordance with the requirements of Section R703.4.
4. EIFS shall be installed in accordance with the manufacturer's instructions.
5. EIFS shall terminate not less than 6 inches (152 mm) above the finished ground level.
6. Decorative trim shall not be face-nailed through the EIFS.

R703.9.2 Exterior Insulation and Finish System (EIFS) With Drainage

EIFS with drainage shall comply with the following:

1. ASTM E2568.
2. EIFS with drainage shall be required over all wall assemblies with the exception of substrates of concrete or masonry wall assemblies.
3. EIFS with drainage shall have an average minimum drainage efficiency of 90 percent when tested in accordance with ASTM E2273.
4. The water-resistive barrier shall comply with Section R703.2 or ASTM E2570.
5. The water-resistive barrier shall be applied between the EIFS and the wall sheathing.
6. Flashing of EIFS with drainage shall be provided in accordance with the requirements of Section R703.4.
7. EIFS with drainage shall be installed in accordance with the manufacturer's instructions.
8. EIFS with drainage shall terminate not less than 6 inches (152 mm) above the finished ground level.
9. Decorative trim shall not be face-nailed through the EIFS with drainage.

R703.10 Fiber Cement Siding

R703.10.1 Panel Siding

Fiber-cement panels shall comply with the requirements of ASTM C1186, Type A, minimum Grade II or ISO 8336, Category A, minimum Class 2. Panels shall be installed with the long dimension either parallel or perpendicular to framing. Vertical and horizontal joints shall occur over framing members and shall be protected with caulking, or with battens or flashing, or be vertical or horizontal shiplap, or otherwise designed to comply with Section R703.1. Panel siding shall be installed with fasteners in accordance with Table R703.3(1) or the approved manufacturer's instructions.

R703.10.2 Lap Siding

Fiber-cement lap siding having a maximum width of 12 inches (305 mm) shall comply with the requirements of ASTM C1186, Type A, minimum Grade II or ISO 8336, Category A, minimum Class 2. Lap siding shall be lapped a minimum of $1\frac{1}{4}$ inches (32 mm) and lap siding not having tongue-and-groove end joints shall have the ends protected with caulking, covered with an H-section joint cover, located over a strip of flashing, or shall be designed to comply with Section R703.1. Lap siding courses shall be installed with the fastener heads exposed or concealed, in accordance with Table R703.3(1) or approved manufacturer's instructions.

R703.11 Vinyl Siding

Vinyl siding shall be certified and *labeled* as conforming to the requirements of ASTM D3679 by an *approved* agency.

R703.11.1 Installation

Vinyl siding, insulated vinyl siding and compatible accessories shall be installed in accordance with the *manufacturer's installation instructions*.

R703.11.1.1 Fasteners

Unless specified otherwise by the manufacturer's instructions, fasteners for vinyl siding shall be 0.120-inch (3 mm) shank diameter nail with a 0.313-inch (8 mm) head.

R703.11.1.2 Penetration Depth

Unless specified otherwise by the manufacturer's installation instructions or in accordance with Table R703.3(1), fasteners shall penetrate into building framing. The total

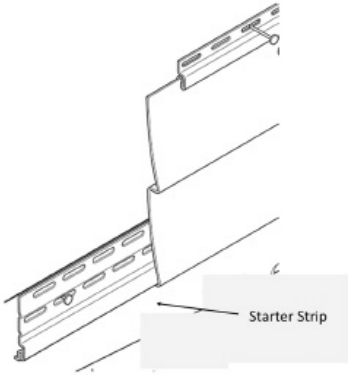
penetration into the sheathing, furring framing or other nailable substrate shall be a minimum 1¼ inches (32 mm).

R703.11.1.3 Spacing

Unless specified otherwise by the manufacturer's installation instructions, the maximum spacing between fasteners for horizontal siding shall be 16 inches (406 mm), and for vertical siding 12 inches (305 mm) both horizontally and vertically. Where specified by the manufacturer's instructions and supported by a test report, alternative fastener spacing is permitted.

R703.11.1.4 Starter Strip

The first course of horizontal siding shall be secured using a starter strip as specified in the manufacturer's installation instructions. See Figure R703.11.1.4(1).Where the first course of siding has to be cut or trimmed, the bottom edge shall be secured with utility trim and snap locks as specified by the manufacturer's installation instructions.

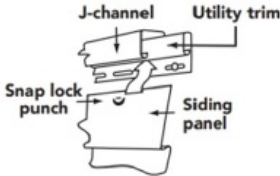


a. Figure R703.11.1.4(1) illustrates typical installation details. See manufacturer's installation instructions for actual installation details.

FIGURE R703.11.1.4(1)
TYPICAL STARTER STRIP^a

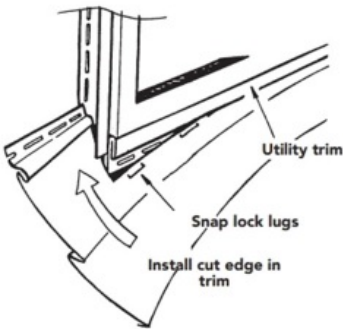
R703.11.1.5 Utility Trim

Where horizontal siding has to be cut or trimmed below windows and at the top of walls, the top edge of the siding shall be secured with utility trim and snap locks or as specified by the manufacturer's installation instructions. See Figures R703.11.1.5(1) and R703.11.5(2).



a. Figure R703.11.1.5(1) illustrates typical installation details. See manufacturer's installation instructions for actual installation details.

FIGURE R703.11.1.5(1)
TYPICAL SNAP LOCK & UTILITY TRIM^a



a. Figure R703.11.1.5(2) illustrates typical installation details. See manufacturer's installation instructions for actual installation details.

FIGURE R703.11.1.5(2)
SNAP LOCK & UTILITY TRIM UNDER WINDOWS^a

R703.11.2 Installation Over Foam Plastic Sheathing

Where vinyl siding or insulated vinyl siding is installed over foam plastic sheathing, the vinyl siding shall comply with Section R703.11 and shall have a wind load design pressure rating in accordance with Table R703.11.2.

Exceptions:

1. Where the foam plastic sheathing is applied directly over wood structural panels, fiberboard, gypsum sheathing or other *approved* backing capable of independently resisting the design wind pressure, the vinyl siding shall be installed in accordance with Sections R703.11.1 and R703.3.3.
2. Where the vinyl siding manufacturer's product specifications provide an *approved* wind load design pressure rating for installation over foam plastic sheathing, use of this wind load design pressure rating shall be permitted and the siding shall be installed in accordance with the *manufacturer's instructions*.
3. Where the foam plastic sheathing and its attachment has a design wind pressure resistance complying with Sections R316.8 and R301.2.1, the vinyl siding shall be installed in accordance with Sections R703.11.1 and R703.3.3.

TABLE R703.11.2

REQUIRED MINIMUM WIND LOAD DESIGN PRESSURE RATING FOR VINYL SIDING INSTALLED OVER FOAM PLASTIC SHEATHING ALONE

Ultimate Design Wind Speed (mph)	Adjusted Minimum Design Wind Pressure (ASD) (psf) ^{a, b}					
	Case 1: With interior gypsum wallboard ^c			Case 2: Without interior gypsum wallboard ^c		
	Exposure			Exposure		
	B	C	D	B	C	D
≤ 95	-30.0	-33.2	-39.4	-33.9	-47.4	-56.2
100	-30.0	-36.8	-43.6	-37.2	-52.5	-62.2
105	-30.0	-40.5	-48.1	-41.1	-57.9	-68.6
110	-31.8	-44.5	-52.8	-45.4	-63.5	-75.3
115	-35.5	-49.7	-59.0	-50.7	-71.0	-84.2
120	-37.4	-52.4	-62.1	-53.4	-74.8	-88.6
130	-44.9	-62.8	-74.5	-64.1	-89.7	-106
> 130	See note d					

For SI: 1 mile per hour = 0.447 m/s, 1 pound per square foot = 0.0479 kPa.

- a. Linear interpolation is permitted.
- b. The table values are based on a maximum 30-ft. mean roof height, an effective wind area of 10 ft², Wall Zone 5 (corner), and the ASD design component and cladding wind pressure from Table R301.2(2), adjusted for exposure in accordance with Table R301.2(3), multiplied by the following adjustment factors: 1.87 (Case 1) and 2.67 (Case 2).
- c. Gypsum wallboard, gypsum panel product or equivalent.
- d. For the indicated wind speed condition and where foam sheathing is the only sheathing on the exterior of a frame wall with vinyl siding, the wall assembly shall be capable of resisting an impact without puncture at least equivalent to that of a wood frame wall with minimum 7/16" OSB sheathing as tested in accordance with ASTM E1886. The vinyl siding shall comply with an adjusted design wind pressure requirement in accordance with note b, using an adjustment factor of 2.67.

R703.11.2.1 Basic Wind Speed Not Exceeding 115 Miles Per Hour and Exposure Category B

Reserved.

R703.11.2.2 Design Wind Pressure Rating

Reserved.

R703.12 Adhered Masonry Veneer Installation

Adhered masonry veneer shall comply with the requirements of Section R703.7.3 and the requirements in Sections 12.1 and 12.3 of TMS 402/ACI 530/ASCE 5. Adhered masonry veneer shall be installed in accordance with Section R703.7.1, Article 3.3C of TMS 602/ACI 530.1/ASCE 6 or the manufacturer's instructions.

R703.12.1 Clearances

On exterior stud walls, adhered masonry veneer shall be installed:

1. Minimum of 4 inches (102 mm) above the earth;
2. Minimum of 2 inches (51 mm) above paved areas; or
3. Minimum of 1/2 inch (12.7 mm) above exterior walking surfaces that are supported by the same foundation that supports the exterior wall.

R703.12.2 Flashing at Foundation

A corrosion-resistant screed or flashing of a minimum 0.019-inch (0.48 mm) or 26-gage galvanized or plastic with a minimum vertical attachment flange of 3½ inches (89 mm) shall be installed to extend a minimum of 1 inch (25 mm) below the foundation plate line on exterior stud walls in accordance with Section R703.4.

R703.12.3 Water-Resistive Barrier

A water-resistive barrier shall be installed as required by Section R703.2 and shall comply with the requirements of Section R703.7.3. The water-resistive barrier shall lap over the exterior of the attachment flange of the screed or flashing provided in accordance with Section R703.7.3.

R703.13 Insulated Vinyl Siding

Insulated vinyl siding shall be certified and labeled as conforming to the requirements of ASTM D7793 by an approved agency.

R703.13.1 Insulated Vinyl Siding and Accessories

Insulated vinyl siding and compatible accessories shall be installed in accordance with Sections R703.11.1, R703.11.2 and the *manufacturer's installation instructions*.

R703.14 Polypropylene Siding

Polypropylene siding shall be certified and labeled as conforming to the requirements of ASTM D7254 by an approved agency. In addition, polypropylene siding shall conform to the fire separation distance requirements of Section R703.14.2 or R703.14.3.

R703.14.1 Polypropylene Siding and Accessories

Polypropylene siding and accessories shall be installed in accordance with manufacturer's installation instructions.

R703.14.1.1 Installation

Unless otherwise specified in the manufacturer's installation instructions, polypropylene siding shall be installed over and attached to wood structural panel sheathing with minimum thickness of 7/16 inch (11.1 mm), or other nailable substrate, composed of wood or wood-based material and fasteners having equivalent withdrawal resistance. Accessories shall be installed in accordance with the manufacturer's installation instructions.

R703.14.1.1.1 Starter Strip

Horizontal siding shall be installed with a starter strip at the initial course at any location.

R703.14.1.1.2 Under Windows and Top of Walls

Where nail hem is removed, such as under windows and at the top of walls, nail slot punch or pre-drilled holes shall be constructed as shown in Figure R703.14.1.1.2(1).

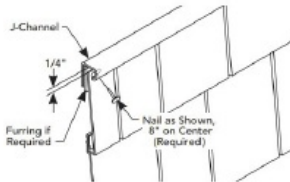


FIGURE R703.14.1.1.2(1)

TRIM UNDER WINDOW AND TOP OF WALLS POLYPROPYLENE SIDING

R703.14.1.2 Fastener Requirements

Unless otherwise specified in the approved manufacturer's installation instructions, nails shall be corrosion resistant, with a minimum 0.120-inch (3 mm) shank and minimum 0.313-inch (8 mm) head diameter. Nails shall be a minimum of 1¼ inches (32 mm) long or as necessary to penetrate sheathing or nailable substrate not less than ¾ inch (19.1 mm). Where the nail fully penetrates the sheathing or nailable substrate, the end of the fastener shall extend not less than ¼ inch (6.4 mm) beyond the opposite face of the sheathing or nailable substrate. Staples are not permitted. The spacing of fasteners shall be in accordance with the manufacturer's installation instructions.

R703.14.2 Fire Separation

Polypropylene siding shall not be installed on walls with a fire separation distance of less than 5 feet (1524 mm) and walls closer than 10 feet (3048 mm) to a building on another lot.

Exception: Walls perpendicular to the line used to determine the fire separation distance.

R703.14.3 Flame Spread Index

The certification of the *flame spread index* shall be accompanied by a test report stating that all portions of the test specimen ahead of the flame front remained in position during the test in accordance with ASTM E84 or UL 723.

R703.15 Cladding Attachment Over Foam Sheathing to Wood Framing

Cladding shall be specified and installed in accordance with Section R703, the cladding manufacturer's approved instructions, including any limitations for use over foam plastic sheathing, or an approved design. In addition, the cladding or furring attachments through foam sheathing to framing shall meet or exceed the minimum fastening requirements of Section R703.15.1, Section R703.15.2, or an approved design for support of cladding weight.

Exceptions:

1. Where the cladding manufacturer has provided approved installation instructions for application over foam sheathing, those requirements shall apply.
2. For exterior insulation and finish systems, refer to Section R703.9.
3. For anchored masonry or stone veneer installed over foam sheathing, refer to Section R703.8.

R703.15.1 Direct Attachment

Where cladding is installed directly over foam sheathing without the use of furring, cladding minimum fastening requirements to support the cladding weight shall be as specified in Table R703.15.1.

TABLE R703.15.1

CLADDING MINIMUM FASTENING REQUIREMENTS FOR DIRECT ATTACHMENT OVER FOAM PLASTIC SHEATHING TO SUPPORT CLADDING WEIGHT^a

CLADDING FASTENER THROUGH FOAM SHEATHING ^b	CLADDING FASTENER TYPE AND MINIMUM SIZE ^{b, c}	CLADDING FASTENER VERTICAL SPACING (inches)	MAXIMUM THICKNESS OF FOAM SHEATHING ^d (inches)									
			16" o.c. Fastener Horizontal Spacing					24" o.c. Fastener Horizontal Spacing				
			Cladding Weight:					Cladding Weight:				
			3 psf	11 psf	15 psf	18 psf	25 psf	3 psf	11 psf	15 psf	18 psf	25 psf
Wood Framing (minimum 1/4-inch penetration)	0.113" diameter nail	6	2.00	1.45	1.00	0.75	DR	2.00	0.85	0.55	DR	DR
		8	2.00	1.00	0.65	DR	DR	2.00	0.55	DR	DR	DR
		12	2.00	0.55	DR	DR	DR	1.8	DR	DR	DR	DR
	0.120" diameter nail	6	3.00	1.70	1.15	0.90	0.55	3.00	1.05	0.65	0.50	DR
		8	3.00	1.20	0.80	0.60	DR	3.00	0.70	DR	DR	DR
		12	3.00	0.70	DR	DR	DR	2.15	DR	DR	DR	DR
	0.131" diameter nail	6	4.00	2.15	1.50	1.20	0.75	4.00	1.35	0.90	0.70	DR
		8	4.00	1.55	1.05	0.80	DR	4.00	0.90	0.55	DR	DR
		12	4.00	0.90	0.55	DR	DR	2.70	0.50	DR	DR	DR
	0.162" diameter nail	6	4.00	3.55	2.50	2.05	1.40	4.00	2.25	1.55	1.25	0.80
		8	4.00	2.55	1.80	1.45	0.95	4.00	1.60	1.10	0.85	0.50
		12	4.00	1.60	1.10	0.85	0.50	4.00	0.95	0.60	DR	DR

For SI: 1 inch = 25.4 mm, 1 pound per square foot = 0.0479 kPa, 1 pound per square inch = 6.895 kPa.

DR = Design required.

o.c. = On center.

- Wood framing shall be Spruce-pine-fir or any wood species with a specific gravity of 0.42 or greater in accordance with AWC NDS.
- The thickness of wood structural panels complying with the specific gravity requirement of note a shall be permitted to be included in satisfying the minimum penetration into framing. For cladding connections to wood structural panels, refer to Table R703.3.2. For brick veneer tie connections to wood structural panels, refer to Table R703.8.4(2).
- Nail fasteners shall comply with ASTM F1667, except nail length shall be permitted to exceed ASTM F1667 standard lengths.
- Foam sheathing shall have a minimum compressive strength of 15 psi in accordance with ASTM C578 or ASTM C1289.

R703.15.2 Furred Cladding Attachment

Where wood furring is used to attach cladding over foam sheathing, furring minimum fastening requirements to support the cladding weight shall be as specified in Table R703.15.2. Where placed horizontally, wood furring shall be preservative-treated wood in accordance with Section R317.1 or naturally durable wood and fasteners shall be corrosion resistant in accordance Section R317.3.

TABLE R703.15.2

FURRING MINIMUM FASTENING REQUIREMENTS FOR APPLICATION OVER FOAM PLASTIC SHEATHING TO SUPPORT CLADDING WEIGHT^{a, b}

FURRING MATERIAL	FRAMING MEMBER	FASTENER TYPE AND MINIMUM SIZE	MINIMUM PENETRATION INTO WALL FRAMING (inches) ^c	FASTENER SPACING IN FURRING (inches)	MAXIMUM THICKNESS OF FOAM SHEATHING ^e (inches)									
					16" o.c. Furring ^f					24" o.c. Furring ^f				
					Siding Weight:					Siding Weight:				
					3 psf	11 psf	15 psf	18 psf	25 psf	3 psf	11 psf	15 psf	18 psf	25 psf
Minimum 1× Wood Furring ^d	Minimum 2× Wood Stud	0.131" diameter nail	1 1/4	8	4.00	2.45	1.75	1.45	0.95	4.00	1.60	1.10	0.85	DR
				12	4.00	1.60	1.10	0.85	DR	4.00	0.95	0.55	DR	DR
				16	4.00	1.10	0.70	DR	DR	3.05	0.60	DR	DR	DR
		0.162" diameter nail	1 1/4	8	4.00	4.00	3.05	2.45	1.60	4.00	2.75	1.85	1.45	0.85
				12	4.00	2.75	1.85	1.45	0.85	4.00	1.65	1.05	0.75	DR
				16	4.00	1.90	1.25	0.95	DR	4.00	1.05	0.60	DR	DR

		No.10 wood screw	1	12	4.00	2.30	1.60	1.20	0.70	4.00	1.40	0.85	0.60	DR
				16	4.00	1.65	1.05	0.75	DR	4.00	0.90	DR	DR	DR
				24	4.00	0.90	DR	DR	DR	2.85	DR	DF	DR	DR
		1/4" lag screw	1 1/2	12	4.00	2.65	1.90	1.50	0.90	4.00	1.65	1.05	0.80	DR
				16	4.00	1.95	1.25	0.95	0.50	4.00	1.10	0.65	DR	DR
				24	4.00	1.10	0.65	DR	DR	3.25	0.50	DR	DR	DR

For SI: 1 inch = 25.4 mm, 1 pound per square foot = 0.0479 kPa, 1 pound per square inch = 6.895 kPa.

DR = Design required.

o.c. = On center

- Wood framing and furring shall be Spruce-pine-fir or any wood species with a specific gravity of 0.42 or greater in accordance with AWC NDS.
- Nail fasteners shall comply with ASTM F1667, except nail length shall be permitted to exceed ASTM F1667 standard lengths.
- The thickness of wood structural panels complying with the specific gravity requirement of note a shall be permitted to be included in satisfying the minimum required penetration into framing.
- Where the required cladding fastener penetration into wood material exceeds $\frac{3}{4}$ inch and is not more than $1\frac{1}{2}$ inches, a minimum 2 × wood furring or an approved design shall be used.
- Foam sheathing shall have a minimum compressive strength of 15 psi in accordance with ASTM C578 or ASTM C1289.
- Furring shall be spaced not more than 24 inches on center, in a vertical or horizontal orientation. In a vertical orientation, furring shall be located over wall studs and attached with the required fastener spacing. In a horizontal orientation, the indicated 8-inch and 12-inch fastener spacing in furring shall be achieved by use of two fasteners into studs at 16 inches and 24 inches on center, respectively.

R703.16 Cladding Attachment Over Foam Sheathing to Cold-Formed Steel Framing

Cladding shall be specified and installed in accordance with Section R703, the cladding manufacturer's approved instructions, including any limitations for use over foam plastic sheathing, or an approved design. In addition, the cladding or furring attachments through foam sheathing to framing shall meet or exceed the minimum fastening requirements of Section R703.16.1, Section R703.16.2 or an approved design for support of cladding weight.

Exceptions:

- Where the cladding manufacturer has provided approved installation instructions for application over foam sheathing, those requirements shall apply.
- For exterior insulation and finish systems, refer to Section R703.9.
- For anchored masonry or stone veneer installed over foam sheathing, refer to Section R703.7.

R703.16.1 Direct Attachment

Where cladding is installed directly over foam sheathing without the use of furring, cladding minimum fastening requirements to support the cladding weight shall be as specified in Table R703.16.1.

TABLE R703.16.1

CLADDING MINIMUM FASTENING REQUIREMENTS FOR DIRECT ATTACHMENT OVER FOAM PLASTIC SHEATHING TO SUPPORT CLADDING WEIGHT^{a, b}

CLADDING FASTENER THROUGH FOAM SHEATHING INTO:	CLADDING FASTENER TYPE AND MINIMUM SIZE ^c	CLADDING FASTENER VERTICAL SPACING (inches)	MAXIMUM THICKNESS OF FOAM SHEATHING ^d (inches)									
			16" o.c. Fastener Horizontal Spacing					24" o.c. Fastener Horizontal Spacing				
			Cladding Weight:					Cladding Weight:				
			3 psf	11 psf	15 psf	18 psf	25 psf	3 psf	11 psf	15 psf	18 psf	25 psf
Steel Framing (minimum penetration of steel thickness + 3 threads)	No. 8 screw into 33 mil steel or thicker	6	3.00	2.95	2.50	2.20	1.45	3.00	2.35	1.75	1.25	DR
		8	3.00	2.55	2.00	1.60	0.60	3.00	1.80	0.90	DR	DR
		12	3.00	1.80	0.95	DR	DR	3.00	0.65	DR	DR	DR
	No. 10 screw into 33 mil steel	6	4.00	3.50	3.05	2.70	1.95	4.00	2.90	2.20	1.70	0.55
		8	4.00	3.10	2.50	2.05	1.00	4.00	2.25	1.35	0.70	DR
		12	4.00	2.25	1.35	0.70	DR	3.70	1.05	DR	DR	DR
	No. 10 screw into 43 mil steel or thicker	6	4.00	4.00	4.00	4.00	3.60	4.00	4.00	3.80	3.45	2.70
		8	4.00	4.00	4.00	3.70	3.00	4.00	3.85	3.25	2.80	1.80

		12	4.00	3.85	3.25	2.80	1.80	4.00	3.05	2.15	1.50	DR
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For SI: 1 inch = 25.4 mm, 1 mil = 0.0254 mm, 1 pound per square foot = 0.0479 kPa, 1 pound per square inch = 6.895 kPa.

DR = Design required.

o.c. = On center.

- Steel framing shall be minimum 33 ksi steel for 33 mil and 43 mil steel, and 50 ksi steel for 54 mil steel or thicker.
- Where cladding is attached to wood structural panel sheathing only, fastening requirements shall be in accordance with Table R703.3.2. Where brick veneer ties are attached to wood structural panel sheathing only, fastening requirements shall be in accordance with Table R703.8.4(2).
- Screws shall comply with the requirements of ASTM C1513.
- Foam sheathing shall have a minimum compressive strength of 15 psi in accordance with ASTM C578 or ASTM C1289.

R703.16.2 Furred Cladding Attachment

Where steel or wood furring is used to attach cladding over foam sheathing, furring minimum fastening requirements to support the cladding weight shall be as specified in Table R703.16.2. Where placed horizontally, wood furring shall be preservative-treated wood in accordance with Section R317.1 or naturally durable wood and fasteners shall be corrosion resistant in accordance with Section R317.3. Steel furring shall have a minimum G60 galvanized coating.

TABLE R703.16.2

FURRING MINIMUM FASTENING REQUIREMENTS FOR APPLICATION OVER FOAM PLASTIC SHEATHING TO SUPPORT CLADDING WEIGHT^a

FURRING MATERIAL	FRAMING MEMBER	FASTENER TYPE AND MINIMUM SIZE ^b	MINIMUM PENETRATION INTO WALL FRAMING (inches)	FASTENER SPACING IN FURRING (inches)	MAXIMUM THICKNESS OF FOAM SHEATHING ^d (inches)									
					16" o.c. Furring ^e					24" o.c. Furring ^e				
					Cladding Weight:					Cladding Weight:				
					3 psf	11 psf	15 psf	18 psf	25 psf	3 psf	11 psf	15 psf	18 psf	25 psf
Minimum 33 mil Steel Furring or Minimum 1x Wood Furring ^c	33 mil Steel Stud	No. 8 screw	Steel thickness + 3 threads	12	3.00	1.80	0.95	DR	DR	3.00	0.65	DR	DR	DR
				16	3.00	1.00	DR	DR	DR	2.85	DR	DR	DR	DR
				24	2.85	DR	DR	DR	DR	2.20	DR	DR	DR	DR
		No. 10 screw	Steel thickness + 3 threads	12	4.00	2.25	1.35	0.70	DR	3.70	1.05	DR	DR	DR
				16	3.85	1.45	DR	DR	DR	3.40	DR	DR	DR	DR
				24	3.40	DR	DR	DR	DR	2.70	DR	DR	DR	DR
	43 mil or thicker Steel Stud	No. 8 Screw	Steel thickness + 3 threads	12	3.00	1.80	0.95	DR	DR	3.00	0.65	DR	DR	DR
				16	3.00	1.00	DR	DR	DR	2.85	DR	DR	DR	DR
				24	2.85	DR	DR	DR	DR	2.20	DR	DR	DR	DR
		No. 10 screw	Steel thickness + 3 threads	12	4.00	3.85	3.25	2.80	1.80	4.00	3.05	2.15	1.50	DR
				16	4.00	3.30	2.55	1.95	0.60	4.00	2.25	1.05	DR	DR
				24	4.00	2.25	1.05	DR	DR	4.00	0.65	DR	DR	DR

For SI: 1 inch = 25.4 mm, 1 mil = 0.0254 mm, 1 pound per square foot = 0.0479 kPa, 1 pound per square inch = 6.895 kPa.

DR = Design required.

o.c. = On center.

- Wood furring shall be Spruce-pine-fir or any softwood species with a specific gravity of 0.42 or greater. Steel furring shall be minimum 33 ksi steel. Steel studs shall be minimum 33 ksi steel for 33 mil and 43 mil thickness, and 50 ksi steel for 54 mil steel or thicker.
- Screws shall comply with the requirements of ASTM C1513.
- Where the required cladding fastener penetration into wood material exceeds $\frac{3}{4}$ inch and is not more than $1\frac{1}{2}$ inches, a minimum 2-inch nominal wood furring or an approved design shall be used.
- Foam sheathing shall have a minimum compressive strength of 15 psi in accordance with ASTM C578 or ASTM C1289.
- Furring shall be spaced not more than 24 inches (610 mm) on center, in a vertical or horizontal orientation. In a vertical orientation, furring shall be located over wall studs and attached with the required fastener spacing. In a horizontal orientation, the indicated 8-inch and 12-inch fastener spacing in furring shall be achieved by use of two fasteners into studs at 16 inches and 24 inches on center, respectively.

R703.17 Cladding Attachment Over Foam Sheathing to Masonry or Concrete Wall Construction

Cladding shall be specified and installed in accordance with Section 703.3 and the cladding manufacturer's instructions or an approved design. Foam sheathing shall be attached to masonry or concrete construction in accordance with the insulation manufacturer's installation instructions or an approved design. Furring and furring attachments through foam sheathing into concrete or masonry substrate shall be designed to resist design loads determined in accordance with Section R301, including support of cladding weight as applicable. Fasteners used to attach cladding or furring through foam sheathing to masonry or concrete substrates shall be approved for application into masonry or concrete material and shall be installed in accordance with the fastener manufacturer's instructions.

Exceptions:

1. Where the cladding manufacturer has provided approved installation instructions for application over foam sheathing and connection to a masonry or concrete substrate, those requirements shall apply.
2. For exterior insulation and finish systems, refer to Section R703.9.
3. For anchored masonry or stone veneer installed over foam sheathing, refer to Section R703.8.

Section R704 Soffits

R704.1 Wind Resistance of Soffits

Soffits and their attachments shall be capable of resisting wind loads specified in Tables R301.2(2) and R301.2(3) for walls using an effective wind area of 10 square feet.

R704.2 Soffit Installation

Soffit installation shall comply with Sections R704.2.1, R704.2.2, R704.2.3 and R704.2.4.

R704.2.1 Vinyl and Aluminum Soffit Panels

Vinyl and aluminum soffit panels shall be installed using aluminum, galvanized, stainless steel or rust-preventative coated nails or other *approved* corrosion-resistant fasteners specified by the manufacturer and shall be fastened at both ends to a supporting component such as a nailing strip, fascia or subfascia component in accordance with Figure R704.2.1. Where the unsupported span of soffit panels is greater than 12 inches, intermediate nailing strips shall be provided in accordance with Figure R704.2.2 unless a larger span is permitted in accordance with the manufacturer's product approval specification. Vinyl and aluminum soffit panels shall be installed in accordance with the manufacturer's product approval specification and limitations of use. Fascia covers shall be installed in accordance with the manufacturer's product approval specification and limitations of use and Section R704.3.

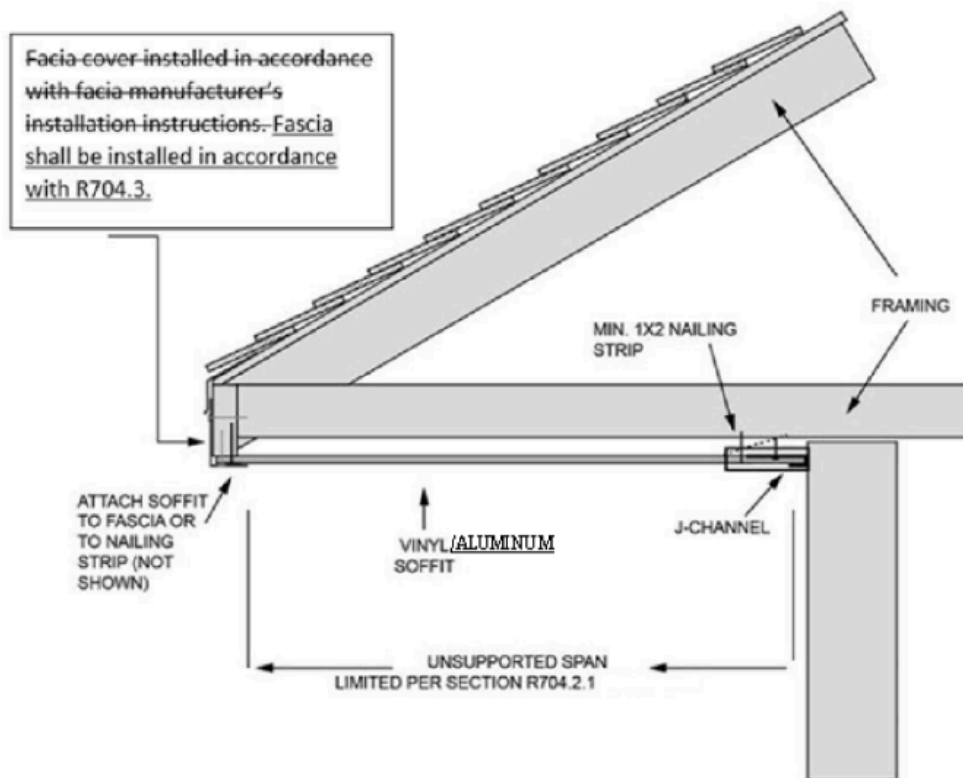


FIGURE 704.2.1(1)

TYPICAL SINGLE-SPAN VINYL OR ALUMINUM SOFFIT PANEL SUPPORT

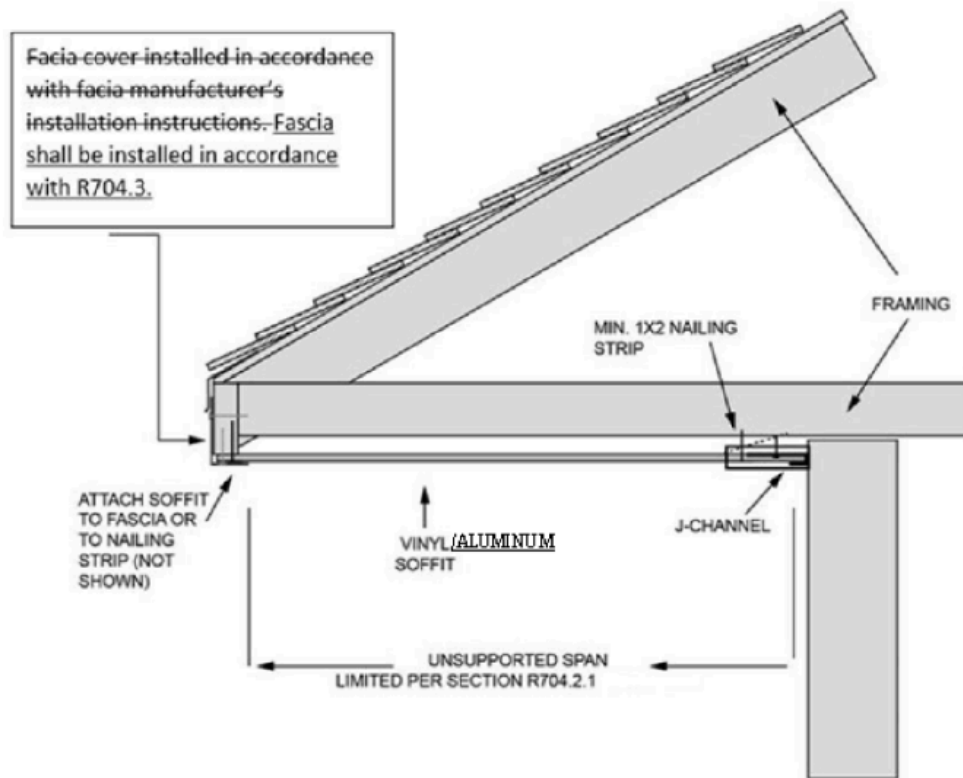


FIGURE 704.2.1(2)

TYPICAL DOUBLE-SPAN VINYL OR ALUMINUM SOFFIT PANEL SUPPORT

R704.2.2 Fiber-Cement Soffit Panels

Fiber-cement soffit panels shall be a minimum of $\frac{1}{4}$ inch thick and shall comply with the requirements of ASTM C1186, Type A, minimum Grade II or ISO 8336, Category A, minimum Class 2. Panel joints shall occur over framing or over wood structural panel sheathing. Soffit panels shall be installed with spans and fasteners in accordance with the manufacturer's product approval specification and limitations of use.

R704.2.3 Hardboard Soffit Panels

Where the design wind pressure is 30 psf or less, soffit panels shall be a minimum of $\frac{7}{16}$ inch in thickness and shall be fastened to framing or nailing strips with $2\frac{1}{2}$ " $\times$ 0.113" siding nails spaced not more than 6 inches on center at panel edges and 12 inches on center at intermediate supports. Where the design wind pressure is greater than 30 psf, hardboard soffit panels shall be installed in accordance with the manufacturer's product approval specification and limitations of use.

R704.2.4 Wood Structural Panel Soffit Prescriptive Alternative

Wood structural panel soffit panels are permitted to be installed in accordance with Table R704.2.4.

TABLE R704.2.4

INSTALLATION REQUIREMENTS FOR WOOD STRUCTURAL PANEL, CLOSED SOFFIT^{b, c, d, e, f}

MAXIMUM DESIGN PRESSURE (- or + psf)	MINIMUM PANEL SPAN RATING	MINIMUM PANEL PERFORMANCE CATEGORY	NAIL TYPE AND SIZE (inches)	FASTENER ^a SPACING ALONG EDGES AND INTERMEDIATE SUPPORTS (inches)	
				GALVANIZED STEEL	STAINLESS STEEL
30	24/0	3/8	6d box (2 $\times$ 0.099 $\times$ 0.266 head diameter)	6 ^f	4
40	24/0	3/8	6d box (2 $\times$ 0.099 $\times$ 0.266 head diameter)	6	4
50	24/0	3/8	6d box (2 $\times$ 0.099 $\times$ 0.266 head diameter)	4	4
			8d common (2 $\frac{1}{2}$ $\times$ 0.131 $\times$ 0.281 head diameter)	6	6
60	24/0	3/8	6d box (2 $\times$ 0.099 $\times$ 0.266 head diameter)	4	3
			8d common (2 $\frac{1}{2}$ $\times$ 0.131 $\times$ 0.281 head diameter)	6	4
70	24/16	7/16	8d common (2 $\frac{1}{2}$ $\times$ 0.131 $\times$ 0.281 head diameter)	4	4
			10d box (3 $\times$ 0.128 $\times$ 0.312 head diameter)	6	4
80	24/16	7/16	8d common (2 $\frac{1}{2}$ $\times$ 0.131 $\times$ 0.281 head diameter)	4	4
			10d box (3 $\times$ 0.128 $\times$ 0.312 head diameter)	6	4

90	32/16	15/32	8d common ($2\frac{1}{2} \times 0.131 \times 0.281$ head diameter)	4	3
			10d box ($3 \times 0.148 \times 0.312$ head diameter)	6	4

- a. Fasteners shall comply with Sections R703.3.2 and R703.3.3.
- b. Maximum spacing of soffit framing members shall not exceed 24 inches.
- c. Wood structural panels shall be of an exterior exposure grade.
- d. Wood structural panels shall be installed with strength axis perpendicular to supports with a minimum of two continuous spans.
- e. Wood structural panels shall be attached to soffit framing members with specific gravity of at least 0.42. Framing members shall be minimum 2 inch by 3 inch nominal with the larger dimension in the cross section aligning with the length of fasteners to provide sufficient embedment depths.
- f. Spacing at intermediate supports is permitted to be 12 inches on center.

R704.3 Aluminum Fascia

Aluminum fascia shall have a minimum thickness of 0.019 inches and be installed in accordance with the manufacturer's instructions and this code. Fasteners shall be aluminum or stainless steel. Aluminum fascia shall be attached in accordance with Section R704.3.1 or R704.3.2 and R704.4 or R704.5. The drip edge shall comply with R905.2.8.5, and the thickness of the drip edge shall be in accordance with Table R903.2.1.

R704.3.1 Fascia Installation Where the Design Wind Pressure Is 30 psf or Less

Where the design wind pressure is 30 pounds per square foot (1.44 kPa) or less, aluminum fascia shall be attached as follows:

1. Finish nails shall be provided in the return leg ($\frac{1}{4}'' \times 0.057'' \times 0.177''$ head diameter) spaced a maximum of 24 inches (610 mm) on center, and
2. The fascia shall be inserted under the drip edge with not less than half the height of the drip edge or 1.0 inch (25 mm), whichever is greater, of the fascia material covered by the drip edge. One finish nail shall be centered in the face of the fascia from each end of the fascia material section located no more than 1 inch below the drip edge.

R704.3.2 Fascia Installation Where the Design Wind Pressure Exceeds 30 psf

Where the design wind pressure is greater than 30 pounds per square foot (1.44 kPa), aluminum fascia shall be attached with one a finish nail [$\frac{1}{4}$ inches by 0.57 inch by 0.177 inch head diameter (32 mm $\times$ 14.5 mm $\times$ 4.5 mm)] in the return leg spaced a maximum of 16 inches (406 mm) on center. The fascia shall be inserted under the drip edge with not less than half the height of the drip edge or 1.0 inch (25 mm), whichever is greater, of the fascia material covered by the drip edge. And one of the following additional attachments:

4. One finish nail shall be centered in the face of the fascia from each end of the fascia material section located no more than 1 inch (25mm) below the drip edge.
5. Top edge of the fascia is secured using utility trim installed beneath the drip edge with snap locks punched into the fascia spaced not more than 6 inches (152 mm) on center, or
6. An approved adhesive applied to the inside of the fascia cover or onto the exterior face of the subfascia framing member.

R704.3.2.1

Where the height of the fascia from the top of the roof sheathing to the bottom of the subfascia plus any thickness of soffit material below the subfascia is less than or equal to 6.5 inches (165 mm) or less, aluminum fascia shall be attached as follows:

1. Finish nails shall be provided in the return leg ($\frac{1}{4}'' \times 0.057'' \times 0.177''$ head diameter) spaced a maximum of 24 inches (610 mm) on center, and
2. The fascia shall be inserted under the drip edge with not less than half the height of the drip edge or 1 inch (25 mm), whichever is greater, of the fascia material covered by the drip edge. One finish nail shall be centered in the face of the fascia from each end of the fascia material section located no more than 1 inch (25 mm) below the drip edge.

R704.3.2.2

Where the height of the fascia from the top of the roof sheathing to the bottom of the sub-fascia plus any thickness of soffit material below the subfascia is greater than 6.5 inches (165 mm), the top edge of the fascia shall be secured using utility trim installed beneath the drip edge with snap locks punched into the fascia spaced no more than 6 inches (152 mm) on center.

R704.3.3 Fascia Installation Where the Design Wind Pressure Exceeds 60 psf

Where the design wind pressure is greater than 60 pounds per square foot (2.88 kPa), aluminum fascia shall be attached as follows in accordance with Section R704.3.3.1 or Section R704.3.3.2.

R704.3.3.1

Where the height of the fascia from the top of the roof sheathing to the bottom of the subfascia plus any thickness of soffit material below the subfascia is less than or equal to 4.5 inches (114 mm) or less, aluminum fascia shall be attached as follows:

1. Finish nails shall be provided in the return leg ($\frac{1}{4}'' \times 0.057'' \times 0.177''$ head diameter) spaced a maximum of 24 inches (610 mm) on center, and
2. The fascia shall be inserted under the drip edge with not less than half the height of the drip edge or 1.0 inch (25 mm), whichever is greater, of the fascia material covered by the drip edge. One finish nail shall be centered in the face of the fascia from each end of the fascia material section located no more than 1 inch (25 mm) below the drip edge.

R704.3.3.2

Where the height of the fascia from the top of the roof sheathing to the bottom of the subfascia plus any thickness of soffit material below the subfascia is greater than 4.5 inches (114 mm), the top edge of the fascia shall be secured using utility trim installed beneath the drip edge with snap locks punched into the fascia spaced no more than 6 inches (152 mm) on center.

mm) on center.

R704.4 Corners on Hip Roofs

Fascia shall be bent around corners and extend at least 12 inches (305 mm) beyond the corner. The next fascia material section shall overlap the extension a minimum of 3 inches (76 mm) and be fastened through the return leg at the overlap.

R704.5 Corners on Gable Roofs

Fascia shall be wrapped (tabbed) around and extend at least 1 inch (25 mm) beyond the corner. The gable fascia material section shall overlap the tab and be fastened through the fascia cover and the tab at the end with two face nails ($1\frac{1}{4}" \times 0.057" \times 0.177"$ head diameter) for a 2 × 4-inch subfascia and three face nails for 2 × 6- inch and greater sub fascia.